CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Schiebinger

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Run

Split	train
Stimulus penalisation	1.0
Prior penalisation	1.0
Genes	100
Stimuli	2
Observed cells	5413
Timepoints	0, 12, 24, 36, 48, 60, 72, 84, 96, 108, 120, 132, 144, 156, 168, 180, 192, 204, 216, 228, 240, 252, 264, 276, 288, 300, 312, 324, 336, 348, 360, 372, 384, 396, 408, 420, 432
Cell types	Neural, MEF, Epithelial, Stromal, Trophoblast, iPSC
Network shown	inter_simul.npy
Embeddings	UMAP; mRNA: log1p=True, normalise=False, cell_depth_for_representation=True (counts / depth factor, model draws taken at the depth of the cells they mimic)

Perturbations (perturbation_simulation)

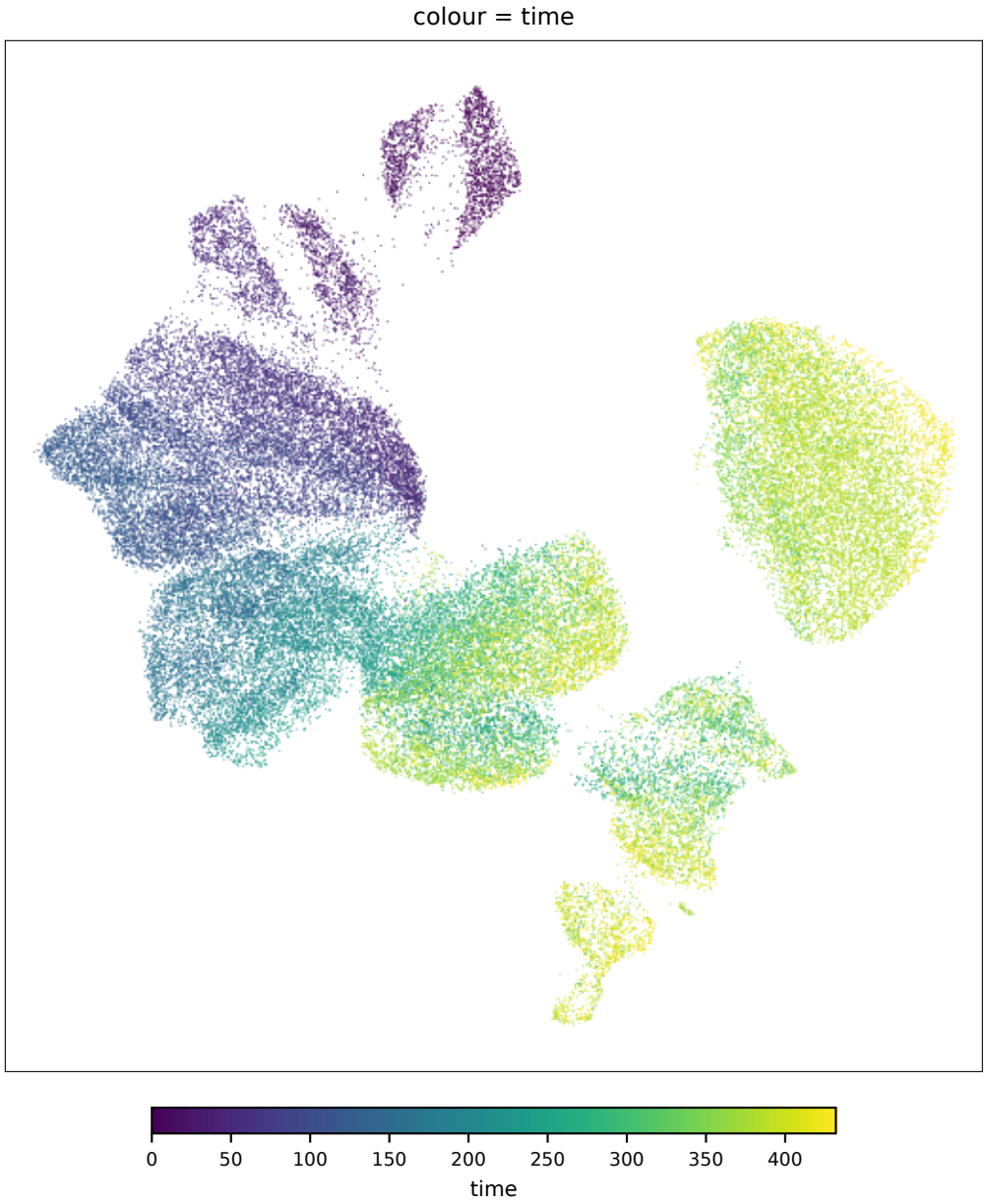
- ✓ P1 OV Obox6
- ✓ P2 OV Zfp42

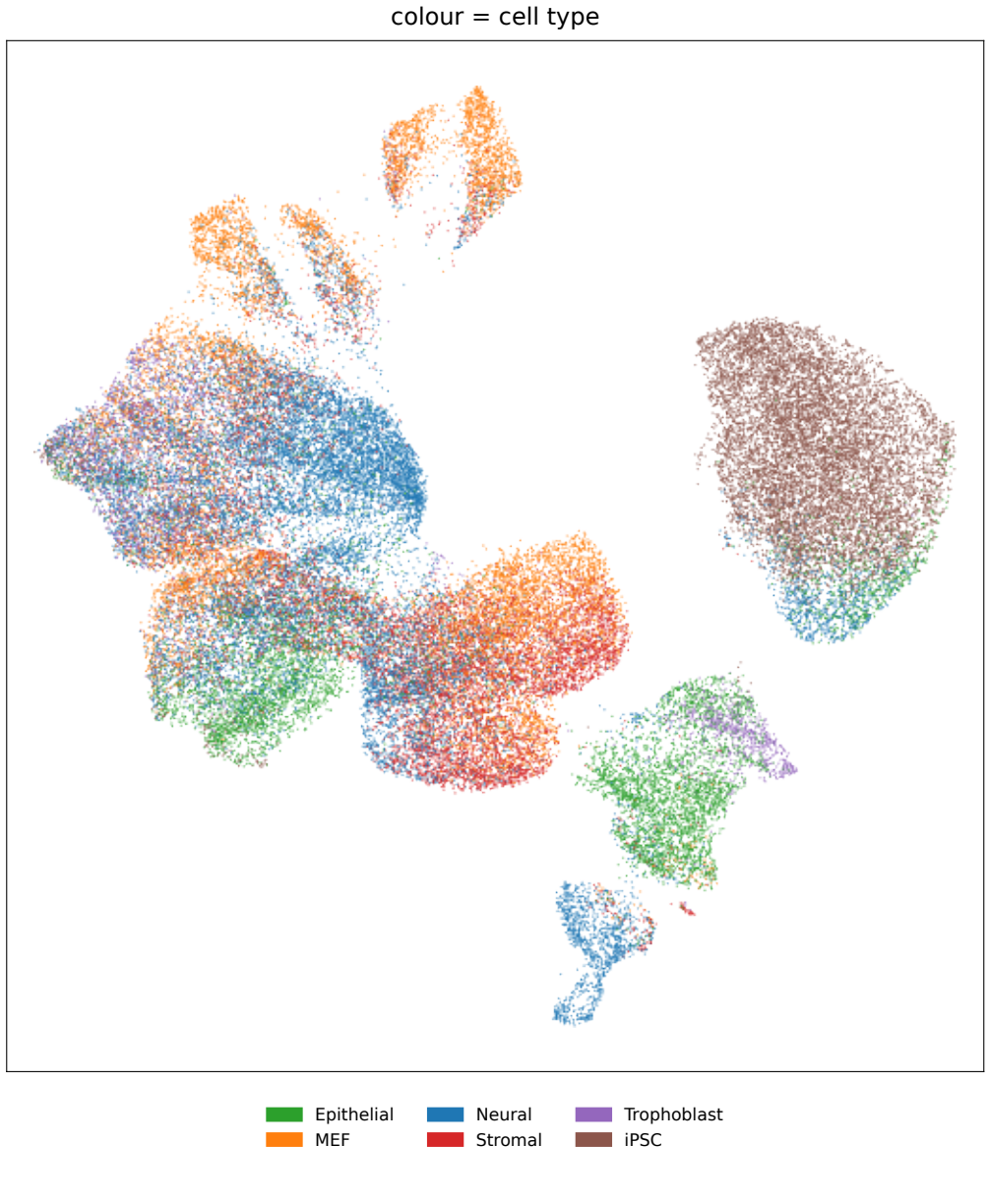
Contents

Data — UMAP of all the cells of Data/data.h5ad (time, samples), also on the genes of the model with select_genes; teaser: classical OT vs CardamomOT displacement fields (
1. Generative model — UMAPs of data, trajectories, NB mixture, network modes and simulation; cell-type proportions; gene-pair correlations; proteins
2. Gene regulatory network — regulatory power (violin plots), top-20 regulators + stimulus and top-20 regulated genes (mutual top-10 edges)
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth; protein dilution at the birth rate
5. Learned dynamics — mRNA and protein fields (mechanistic velocity, displacement along trajectories), summary on mRNA
6. Held-out test cells — predictions with the network fixed vs the test data

Data — UMAP of all the cells

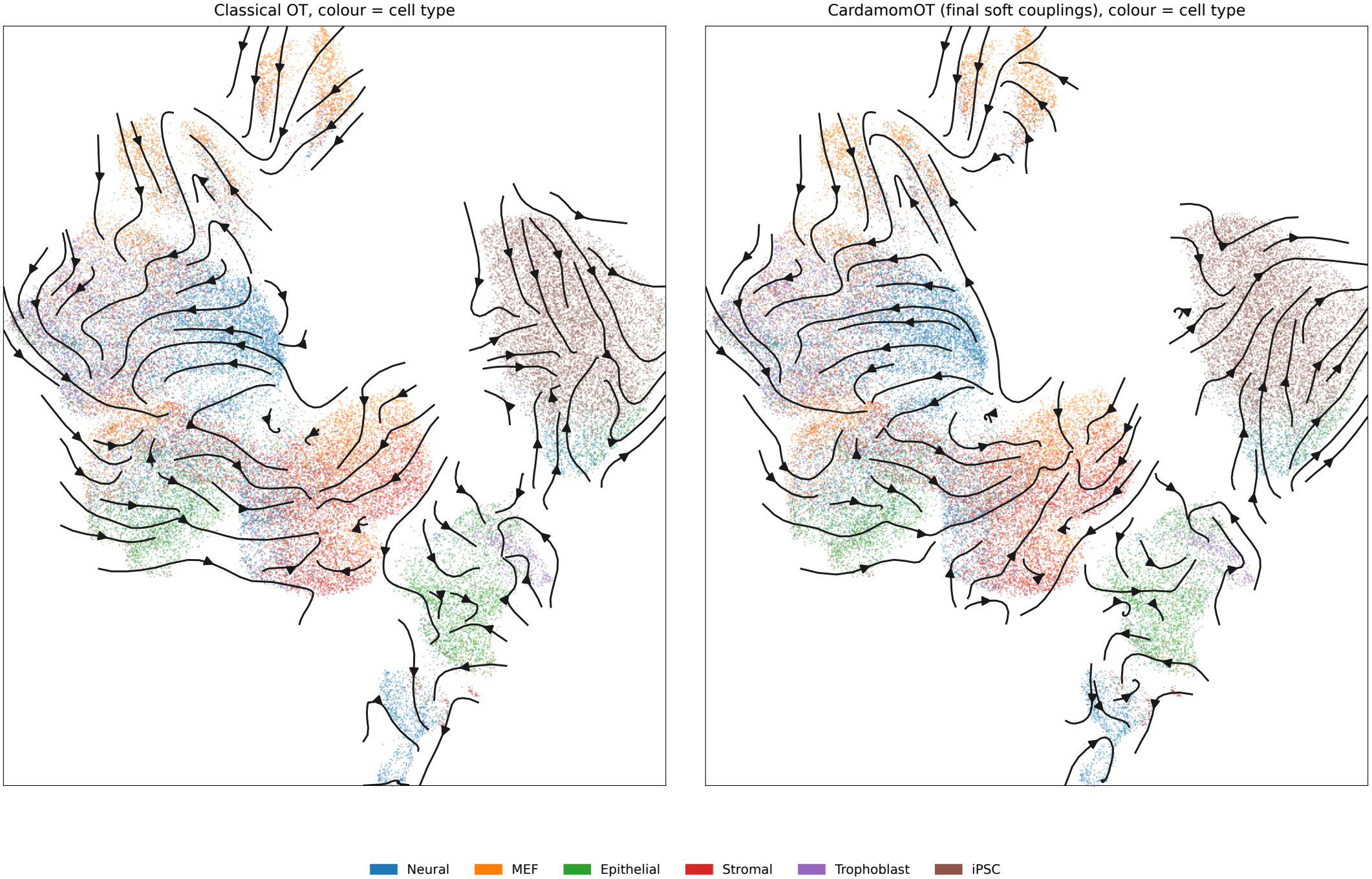
Data/data.h5ad, 50897 cells: log1p(counts), 2000 highly variable genes, PCA (50 components), kNN graph, UMAP. Stored in cardamomOT/embedding_data.npz.





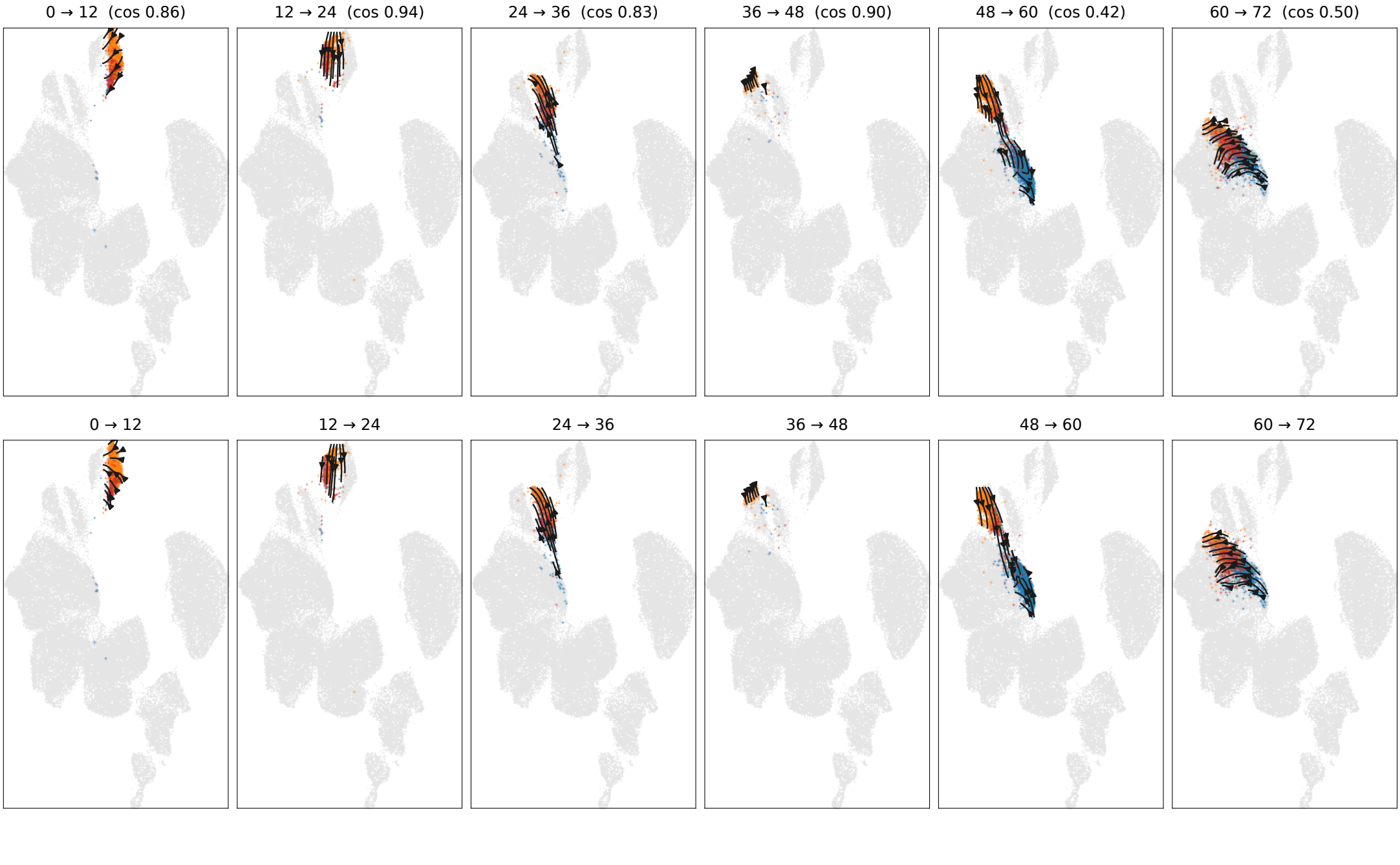
Teaser — displacement fields of classical OT and CardamomOT

Displacement to the mean descendant (x(t+1) − x(t), not divided by Δt), all times. Agreement on the cells with both direct displacements: weighted cosine 0.61 (PCA space). CardamomOT: final soft couplings of the trajectories (5413 cells); cells never reached or outside the training set: Gaussian kernel on the model mRNAs.



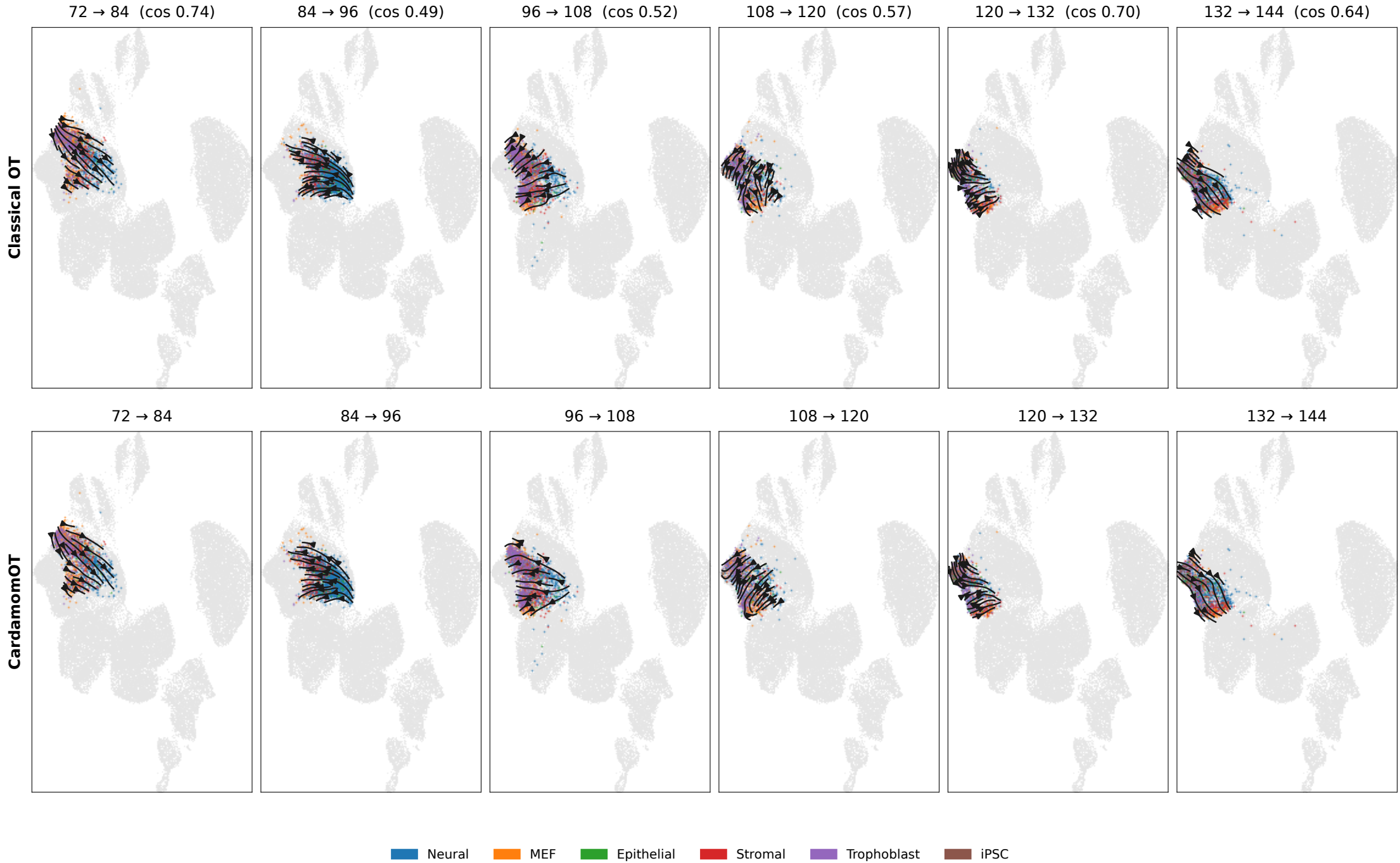
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) of the samples whose next timepoint is t_to, towards their mean descendant at t_to, kNN projection among their cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



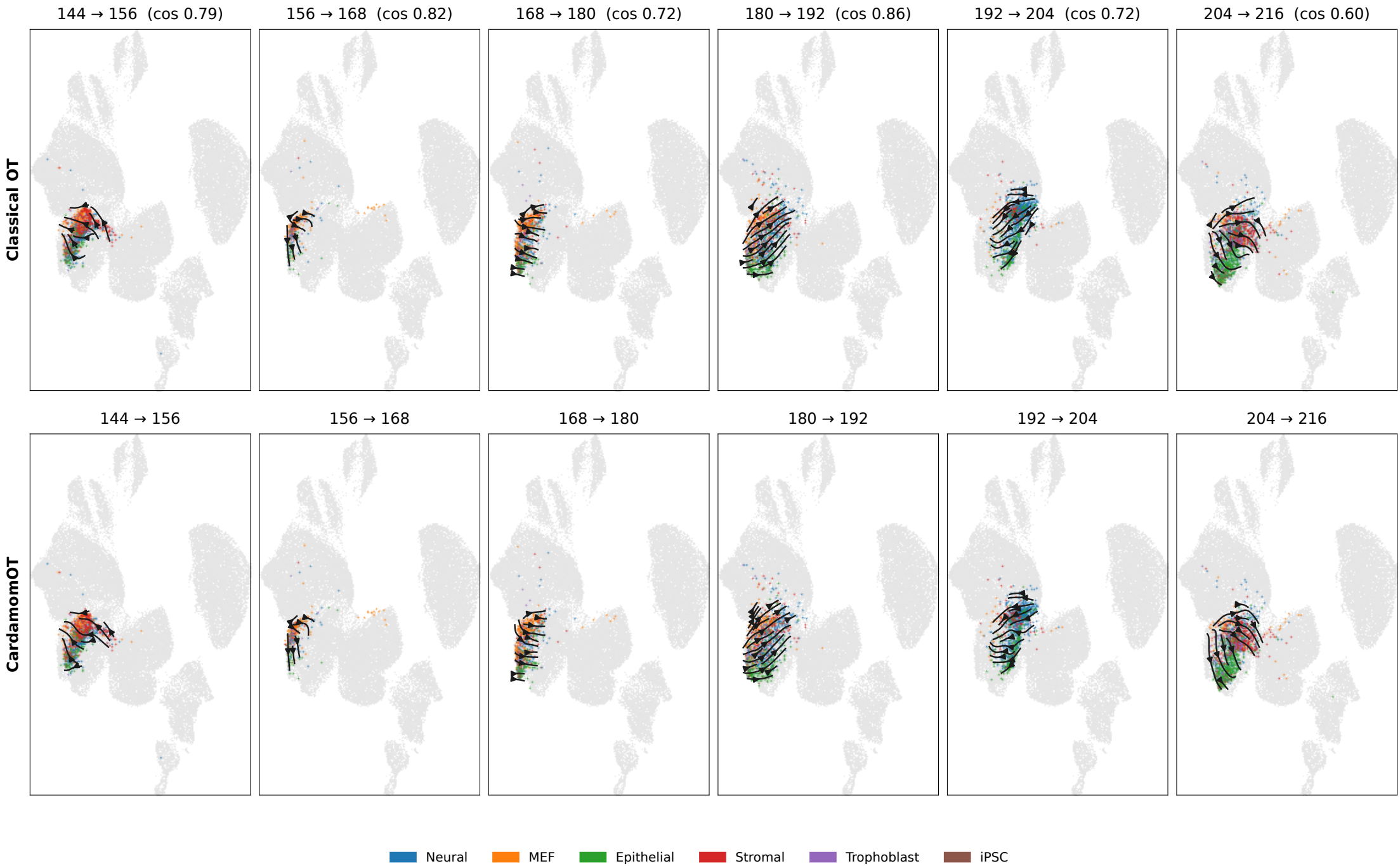
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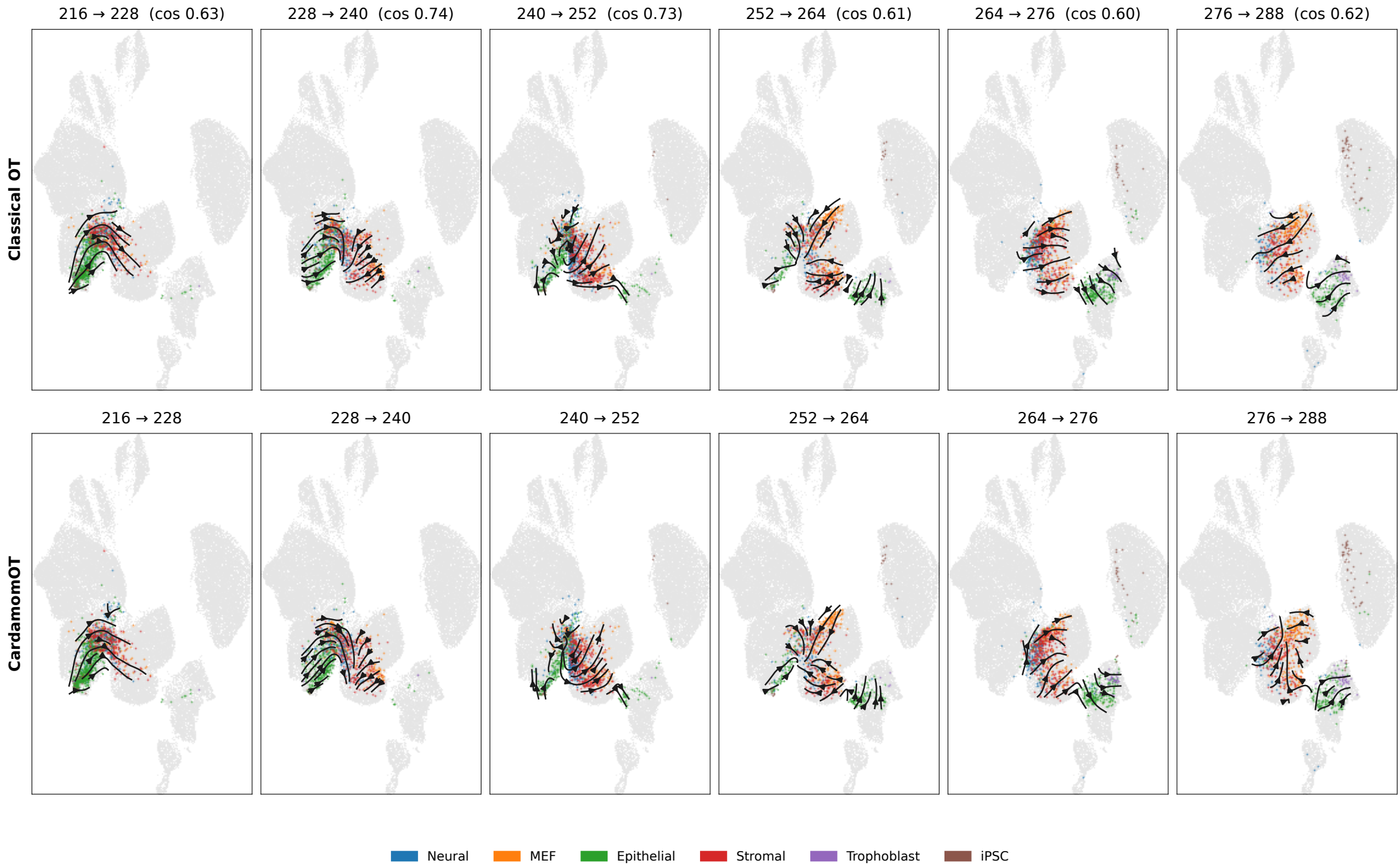
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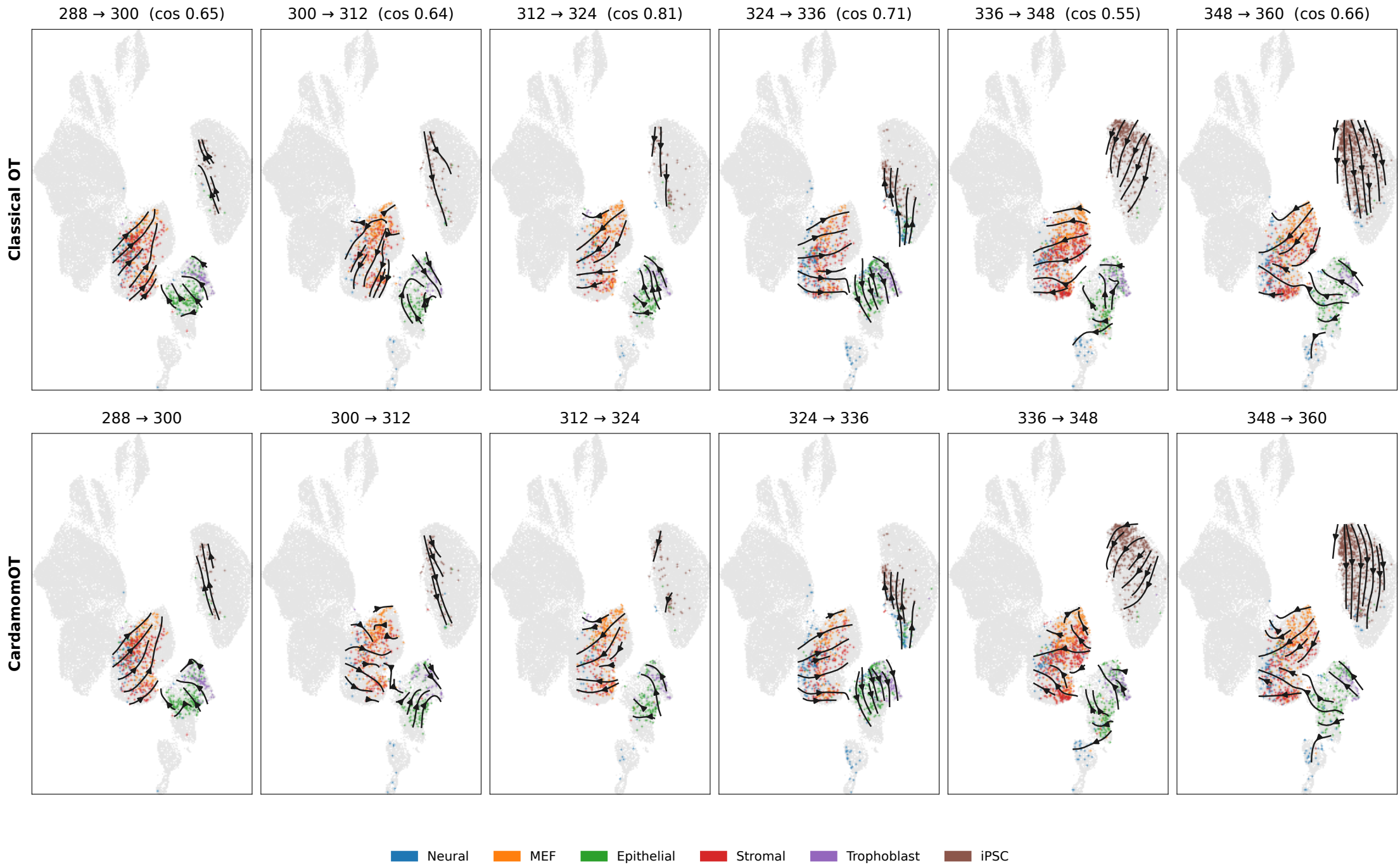
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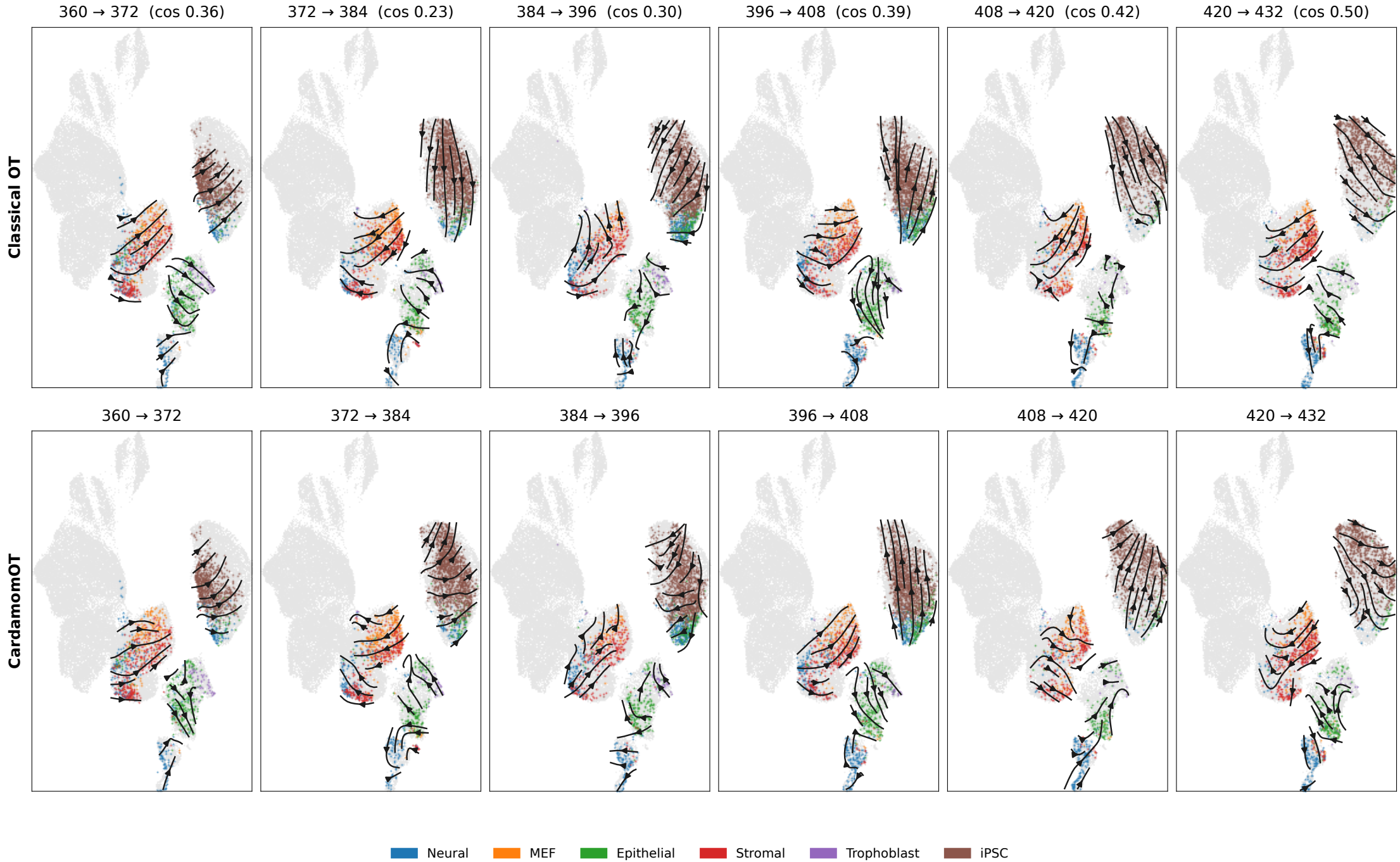
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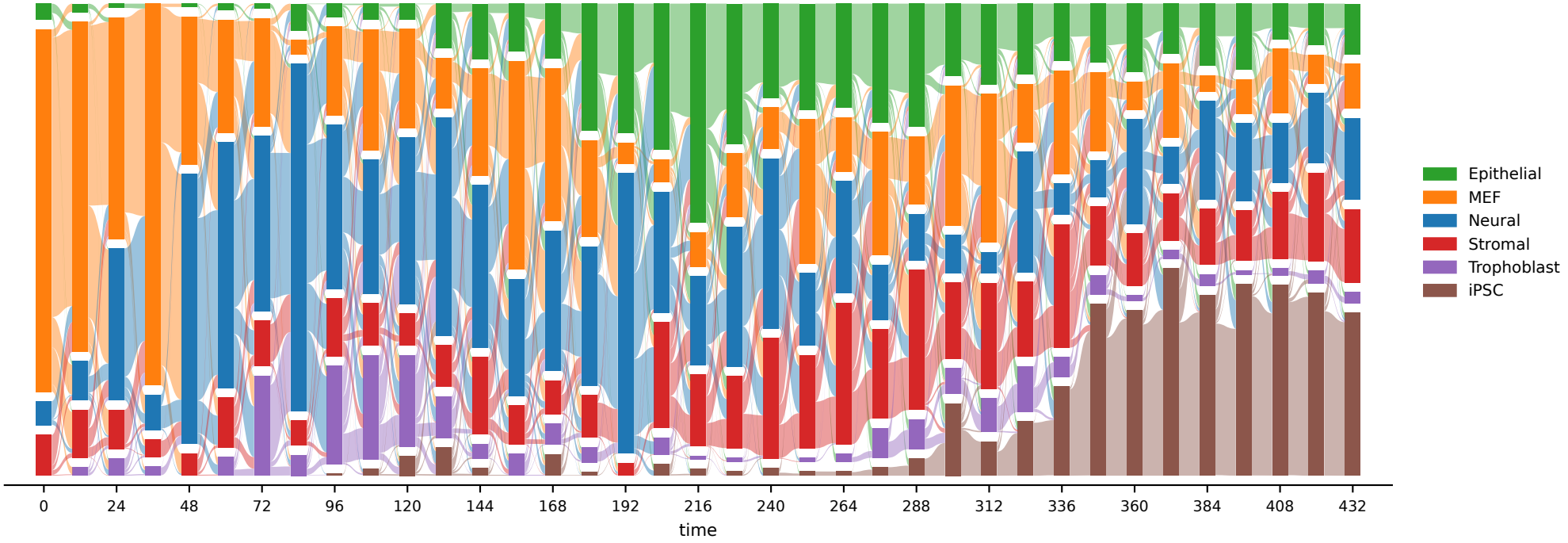
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) of the samples whose next timepoint is t_to, towards their mean descendant at t_to, kNN projection among their cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



Classical OT — cell-type transitions and fate genes

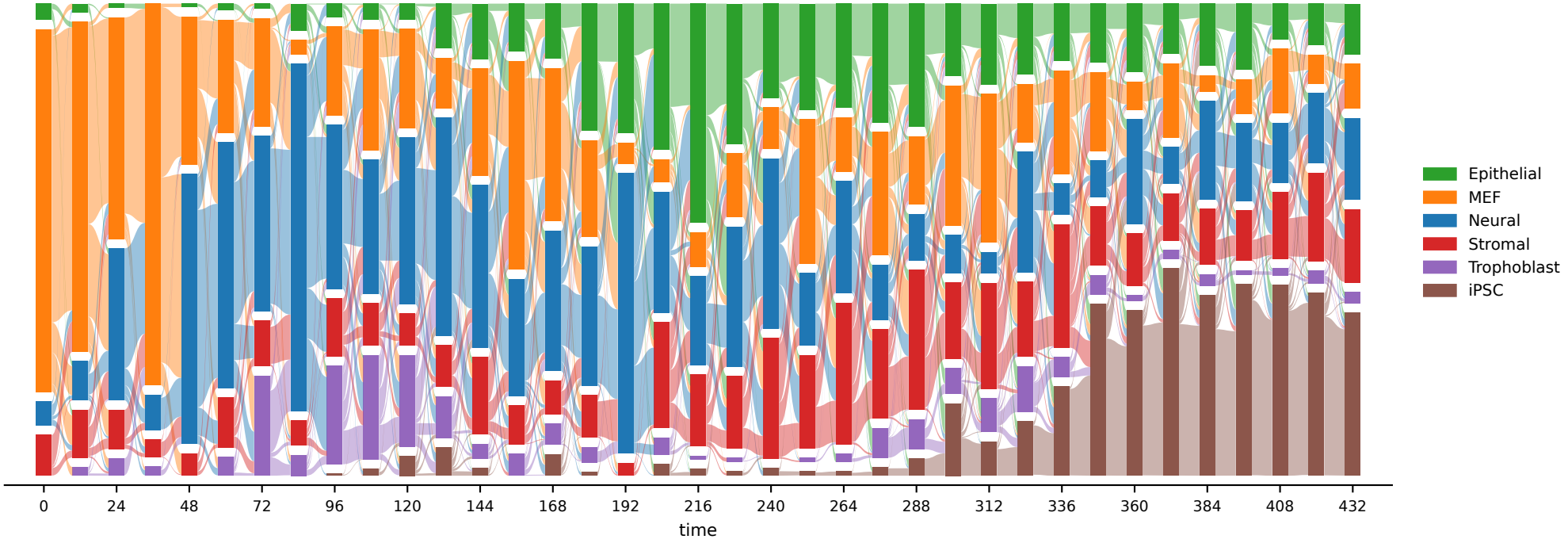
Waddington-OT couplings (every gene, train cells, per sample, growth from the net rate), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (classical_OT/fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
Epithelial	**Krt8** **Krt18** **Epcam** **Krt7** **Dbi** **Bbx** **Mbnl3** Marcksl1 **Xist** **S100a14**	**Krt8** **Krt18** **Epcam** **Krt7** **Bbx** **Mbnl3** **S100a14** **Dbi** **Xist** Hcfc1r1
MEF	**Col1a2** **Thbs1** **Col1a1** **Tagln** **Vim** **Timp2** **Timp3** Sparc **Cd44** Emp3	**Col1a2** **Vim** **Tagln** **Thbs1** **Timp2** **Cd44** Tpm1 **Col1a1** **Timp3** Fbn1
Neural	**Vim** **Ccnd2** **S100a10** **Xist** **Cks1b** **Emp3** **Id3** **Mras** **Pcolce2** Myc	**Vim** **Ccnd2** **S100a10** **Xist** **Id3** **Cks1b** **Pcolce2** **Mras** Cenpa **Emp3**
Stromal	**Mgp** **Sparc** **Col1a2** **Col1a1** **Vim** **Timp2** **Emp3** **Selm** Id3 **Laptm4a**	**Sparc** **Mgp** **Col1a2** **Col1a1** **Vim** **Emp3** **Timp2** **Selm** **Laptm4a** Efemp2
Trophoblast	**Sri** **Dbi** **Hsp90aa1** **Hmgb2** Krt8 Krt18 **Rbbp7** Mbnl3 Cks1b **Hmgn5**	**Dbi** **Sri** **Hsp90aa1** S100a10 **Rbbp7** **Hmgn5** **Hmgb2** Cyb5r3 Taldo1 Cdkn1c
iPSC	**Dppa5a** **Hsp90aa1** **Mybl2** **Ooep** **Nanog** **Nfatc2ip** **Cdc5l** **Sox2** **Gjb3** Zfp42	**Dppa5a** **Hsp90aa1** **Ooep** **Mybl2** **Nanog** **Cdc5l** **Nfatc2ip** **Sox2** **Gjb3** Fmr1nb

Teaser — CardamomOT: cell-type transitions and fate genes

Final soft couplings of the CardamomOT trajectories (model genes, data_<split> cells), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (cardamomOT/teaser_fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
Epithelial	**Krt8** **Epcam** **Krt18** **Krt7** **S100a14** **Mbnl3** **Bbx** Dbi **Gpx2** Dmrtc2	**Krt8** **Epcam** **Krt18** **Krt7** **Mbnl3** **S100a14** **Bbx** Cdkn1c Gpx3 **Gpx2**
MEF	**Col1a2** **Vim** **Tpm1** **Thbs1** **Emp3** **Tagln** Col1a1 **Timp2** **Cd44** Sparc	**Vim** **Tpm1** **Col1a2** Ccnd2 **Cd44** **Emp3** **Tagln** **Thbs1** S100a10 **Timp2**
Neural	**Vim** **Ccnd2** **Dmrtc2** **Myc** **Pcolce2** **Mras** **Cenpa** S100a10 **Cks1b** Sms	**Dmrtc2** **Vim** **Ccnd2** Id3 **Pcolce2** **Mras** **Cenpa** **Myc** **Cks1b** Crabp2
Stromal	**Mgp** **Sparc** **Col1a2** **Col1a1** **Vim** **Timp2** Id3 **Selm** Prrx1 **Emp3**	**Mgp** **Sparc** **Vim** **Col1a2** **Col1a1** **Emp3** **Selm** Xist Laptm4a **Timp2**
Trophoblast	**Sri** **Dbi** **Cks1b** **Hmgb2** **Hsp90aa1** Mbnl3 **Rbbp7** **Hmgn5** Krt8 Krt18	**Dbi** **Sri** **Hsp90aa1** S100a10 **Rbbp7** **Hmgb2** **Hmgn5** Taldo1 **Cks1b** Cyb5r3
iPSC	**Dppa5a** **Hsp90aa1** **Mybl2** **Nanog** **Ooep** **Nfatc2ip** **Sox2** **Zfp42** Gjb3 **Cdc5l**	**Dppa5a** **Hsp90aa1** **Ooep** **Mybl2** **Nanog** **Cdc5l** **Nfatc2ip** **Sox2** Fmr1nb **Zfp42**

Genes of the model (100)

Data/data_train.h5ad, alphabetical order, coloured by selection role. Italics: no edge inside the selection (probability ≥ selection_edge_prob, literature-feasible;
0 genes). Trajectory preservation (classical OT on these genes vs every gene; network otvelo_granger): velocity cosine 0.88 (random genes 0.77), fate JS distance …

Antxr1	Efemp2	Ltbr	S100a10
Anxa3	Ehd2	Marcksl1	S100a14
Batf3	Emp3	Mbnl3	Sall4
Bbx	Epcam	Mgp	Sat1
Bhlhe40	Eva1b	Mras	Sdc4
Btg2	Fbn1	Msmo1	Selm
Ccnb1	Fmr1nb	Mybl2	Sgol1
Ccnd2	Foxo4	Myc	Sms
Cd44	Gata2	Nanog	Sox2
Cd81	Gjb3	Nasp	Sparc
Cdc5l	Gnl3	Nfatc2ip	Sri
Cdkn1c	Gpx2	Nfic	Tagln
Cenpa	Gpx3	Nfix	Taldo1
Cited2	Hcfc1r1	Obox6	Tcl1
Cks1b	Hmgb2	Ooep	Tgif1
Col1a1	Hmgn5	Pcolce2	Thbs1
Col1a2	Hsp90aa1	Phlda2	Timp2
Crabp2	Id3	Prnp	Timp3
Cyb5r3	Ier3	Prrx1	Tmbim1
Dbi	Igfbp3	Pttg1ip	Tnc
Ddit3	Klf4	Rbbp7	Tpm1
Ddx21	Krt18	Rhoc	Tspo
Dmrtc2	Krt7	Rhox6	Vim
Dppa5a	Krt8	Rhox9	Xist
Ebna1bp2	Laptm4a	Rrp9	Zfp42

perturbed (2) query (26) driver (34) entropy (38)

0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as NB(k, c / s_i) (depth model 'mean') when the depth factor s_i is used.

Depth

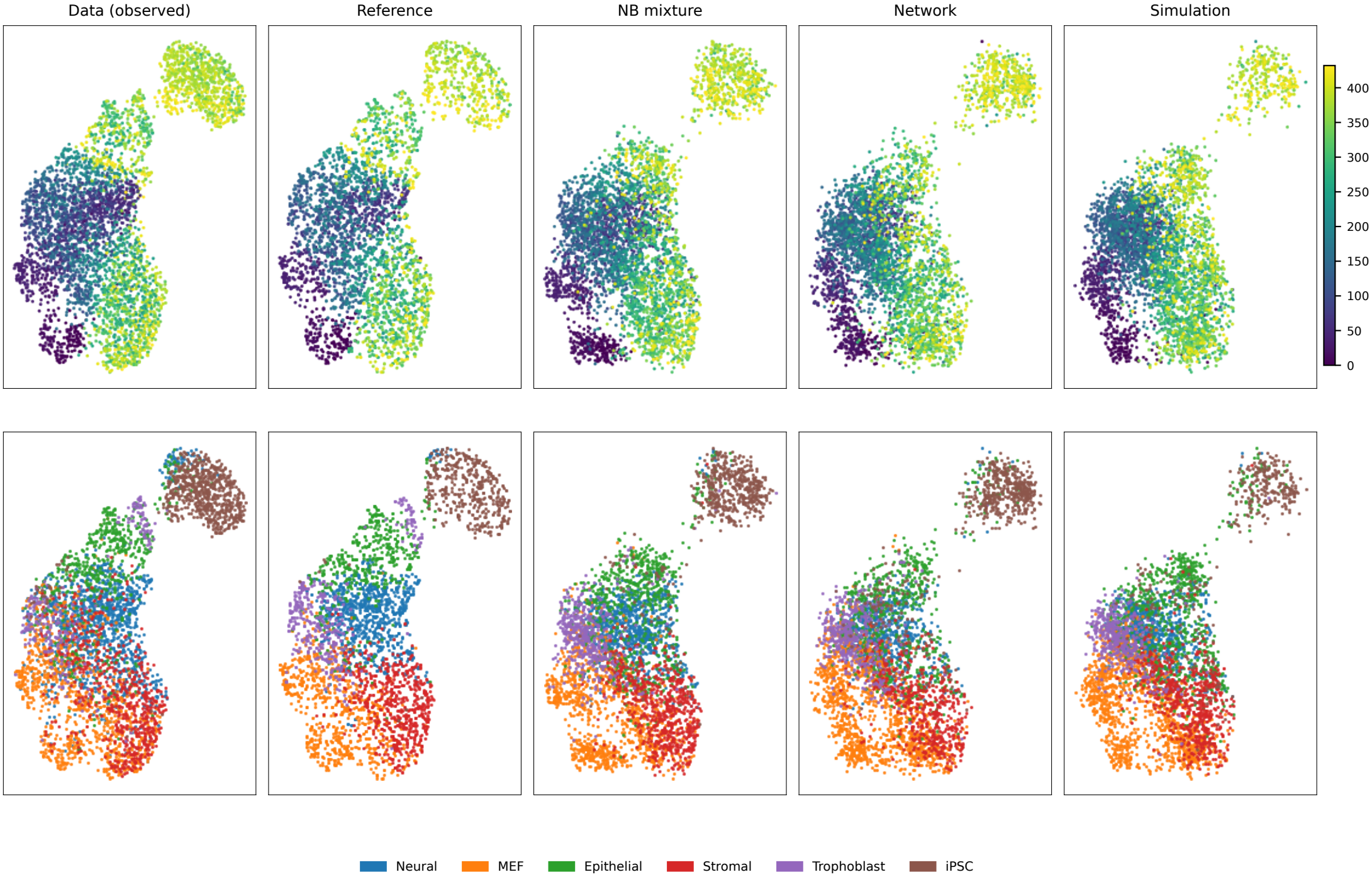
not estimated

Reason

100 genes < 10000

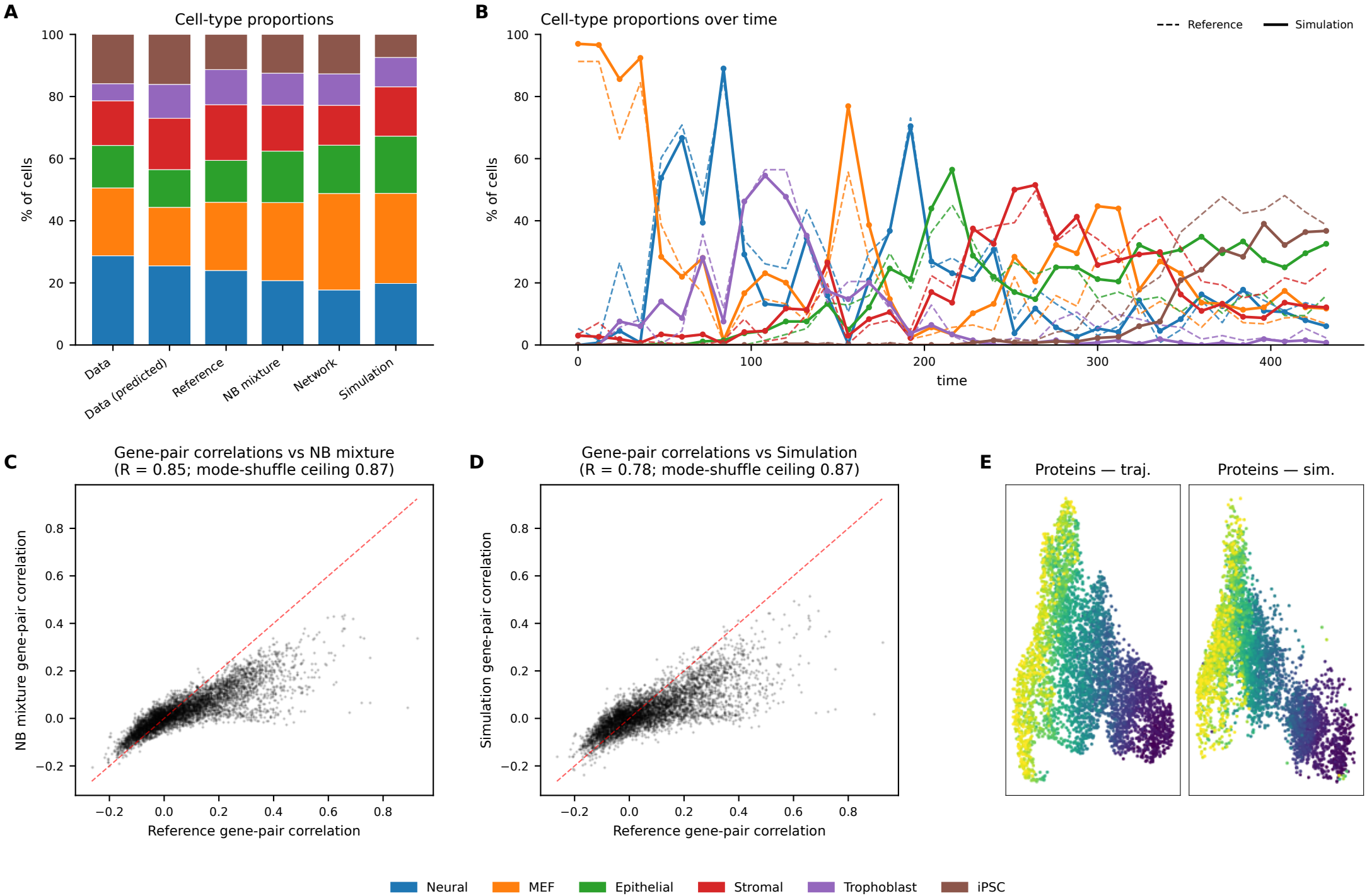
1. Generative model — trajectories and simulation

UMAP learned on the observed data; the trajectories (Reference: states drawn at each time among the observed cells), the NB mixture and network-driven modes along them and the full simulation (without proliferation) are projected onto it.



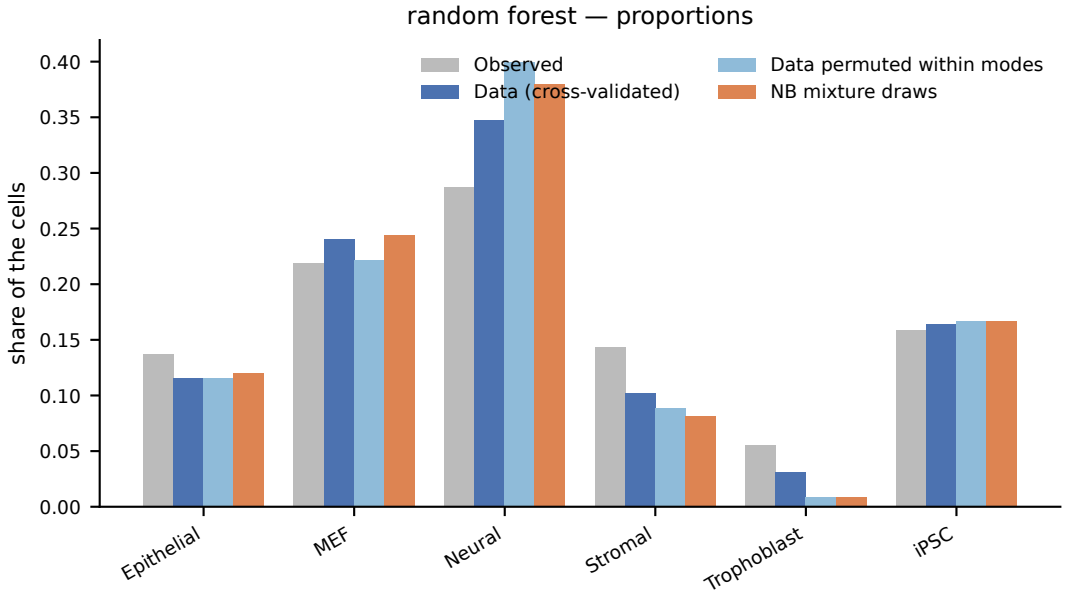
1. Generative model — quantitative checks

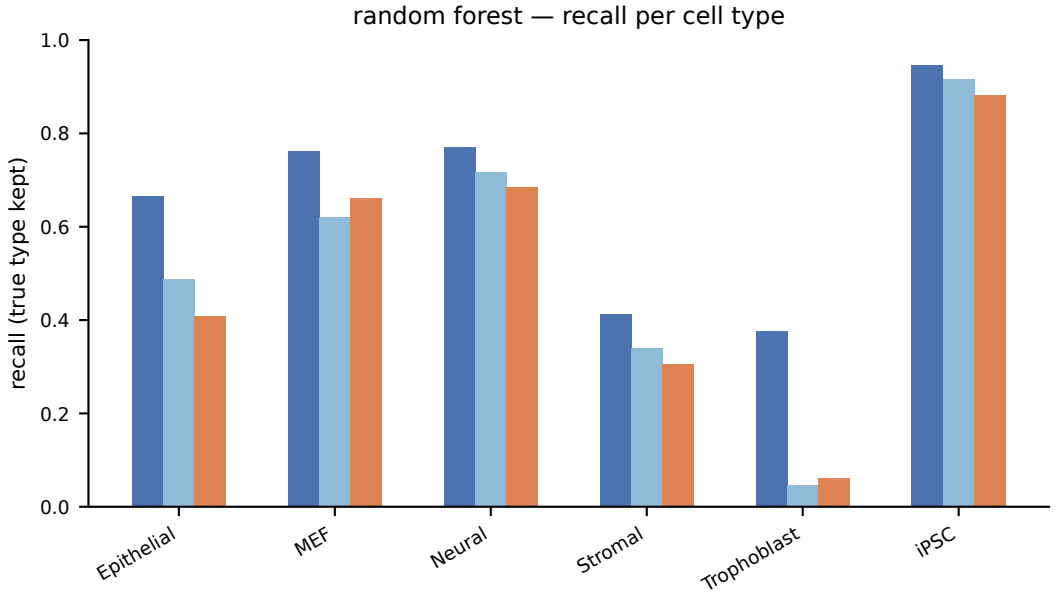
Reference = RNA of the trajectories (states drawn at each time among the observed cells: the observed composition); simulation without proliferation.

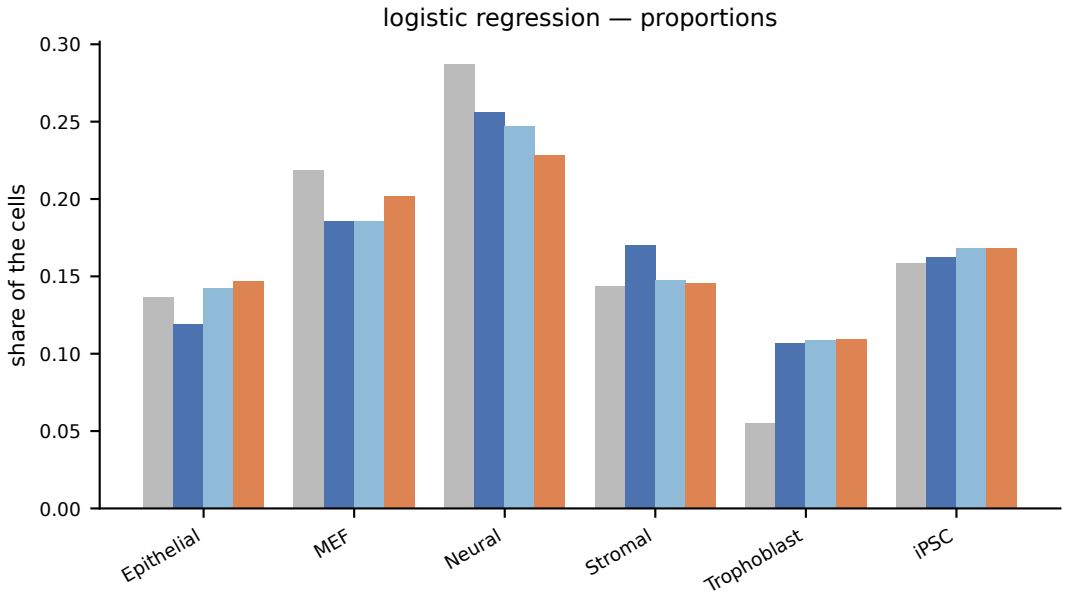


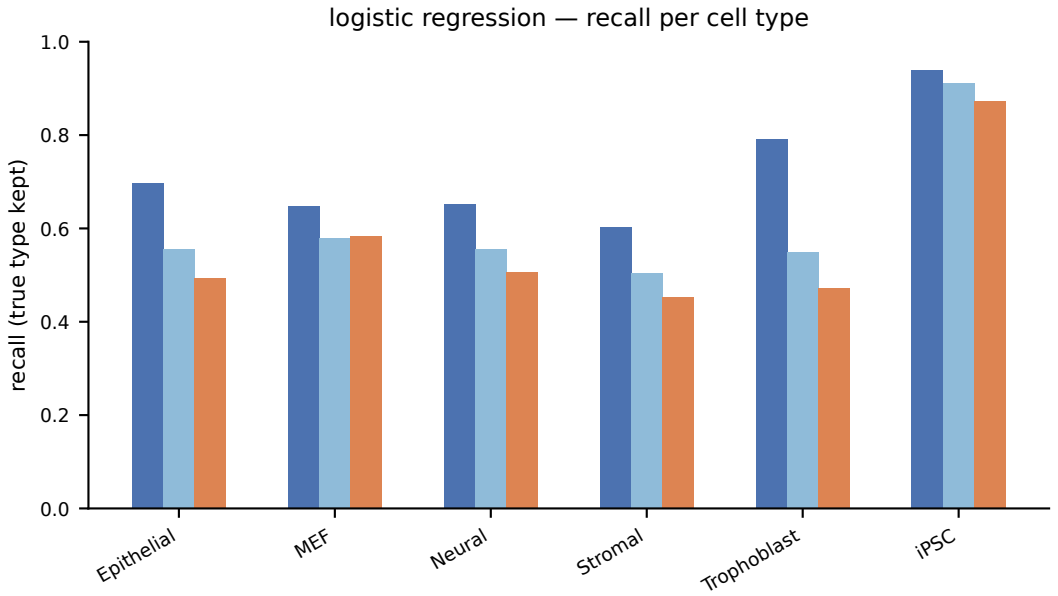
1. Generative model — cell-type identity in the NB mixture

One draw per real cell, per-sample classifiers trained on the other cells. Cross-validated data = classifier limit; permuted within modes (same sample, time, mode) = best model with these modes. Gap data → permuted: modes too coarse; permuted → draws: NB fit.







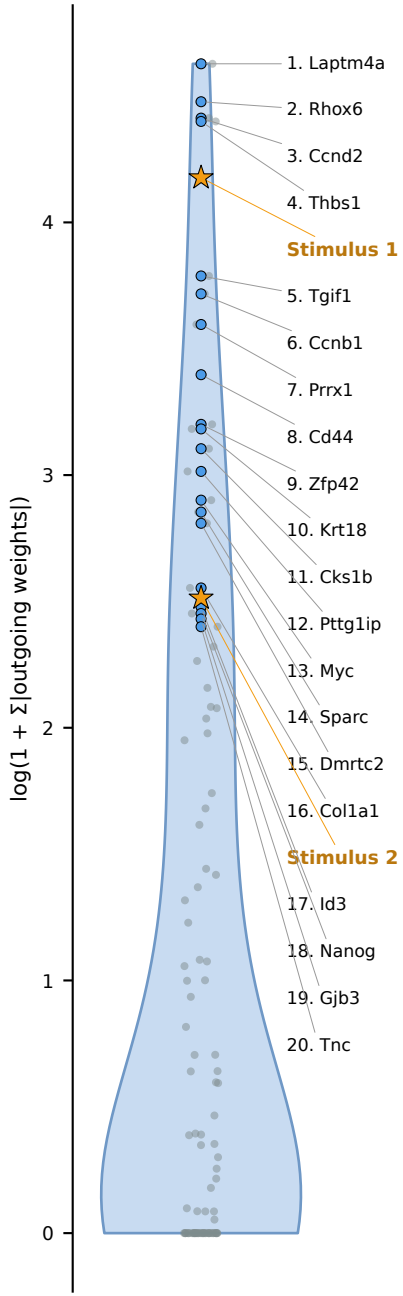


2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 20 genes labelled; stimulus shown as a star (stimulus = 1.0 ≥ 0.5).

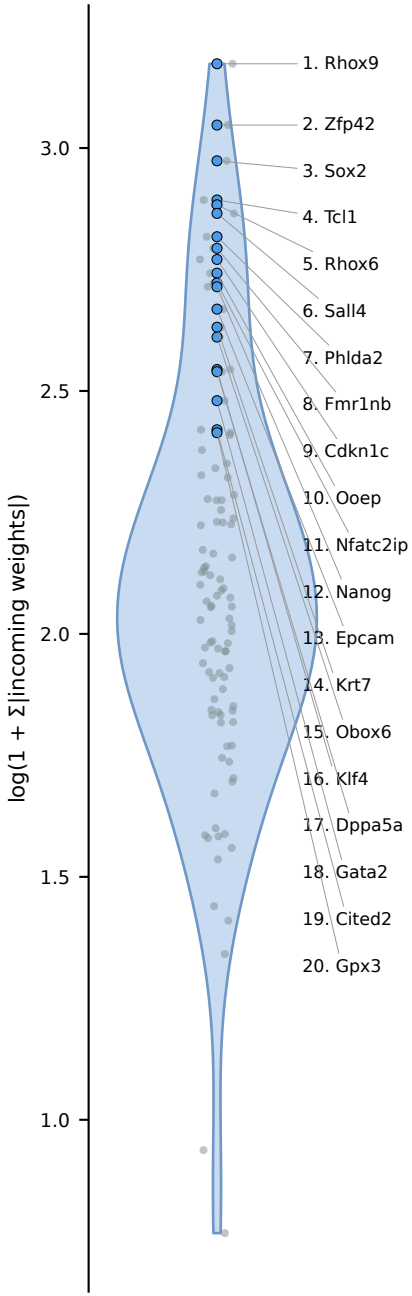
A

Outgoing regulation (regulators)



B

Incoming regulation (targets)



C

Top 20 regulators

#	Regulator	Σ|w|	act.	inh.
1	Laptm4a	101.28	43	2
2	Rhox6	87.02	10	0
3	Ccnd2	81.49	46	15
4	Thbs1	80.41	49	12
5	Tgif1	43.17	39	2
6	Ccnb1	40.15	23	18
7	Prrx1	35.45	33	6
8	Cd44	28.88	18	14
9	Zfp42	23.55	20	2
10	Krt18	23.11	17	0
11	Cks1b	21.30	17	8
12	Pttg1ip	19.38	20	0
13	Myc	17.19	11	12
14	Sparc	16.35	16	0
15	Dmrtc2	15.60	5	4
16	Col1a1	11.85	19	2
17	Id3	10.92	18	6
18	Nanog	10.61	9	2
19	Gjb3	10.36	12	2
20	Tnc	10.02	12	3
★	Stimulus 1	64.15	30	39
★	Stimulus 2	11.35	1	8

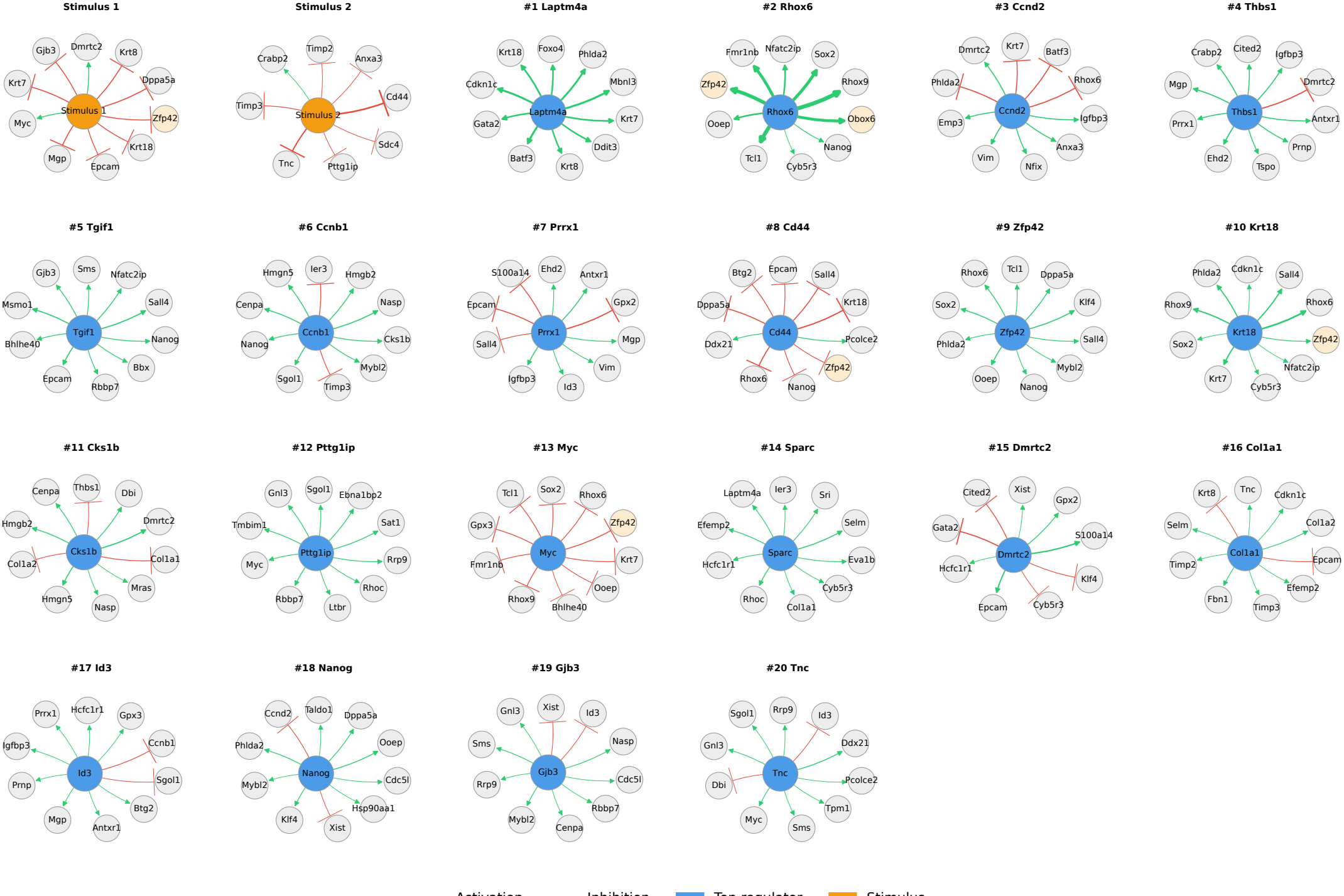
D

Top 20 regulated genes

#	Target	Σ|w|	act.	inh.
1	Rhox9	22.89	3	3
2	Zfp42	20.06	2	4
3	Sox2	18.56	4	4
4	Tcl1	17.05	2	1
5	Rhox6	16.87	6	5
6	Sall4	16.56	6	5
7	Phlda2	15.73	6	2
8	Fmr1nb	15.34	4	1
9	Cdkn1c	14.97	10	3
10	Ooep	14.53	5	4
11	Nfatc2ip	14.21	11	4
12	Nanog	14.10	14	4
13	Epcam	13.42	3	8
14	Krt7	12.89	4	2
15	Obox6	12.61	3	0
16	Klf4	11.73	13	1
17	Dppa5a	11.68	5	3
18	Gata2	10.94	8	1
19	Cited2	10.25	8	1
20	Gpx3	10.18	9	4

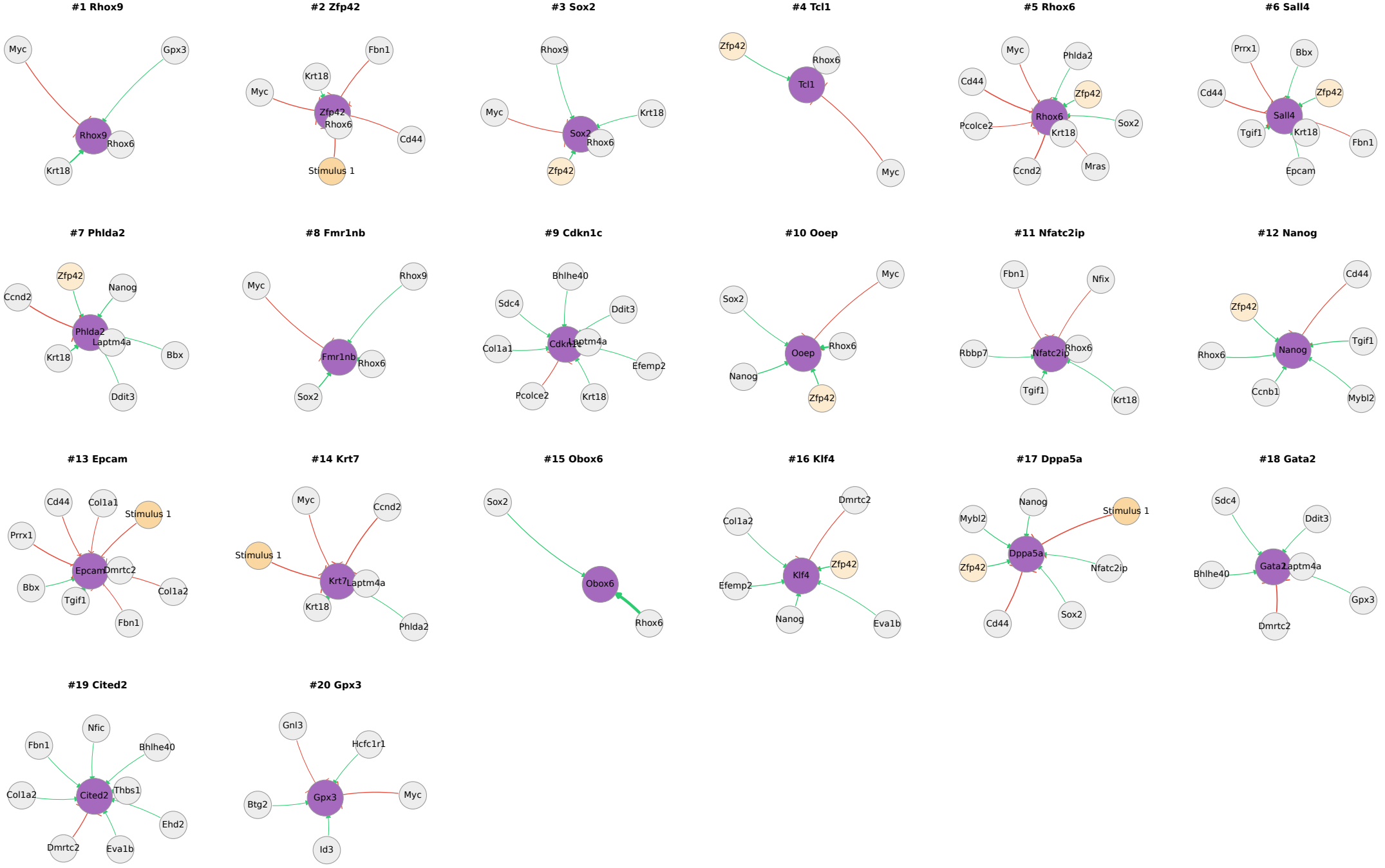
2. Gene regulatory network — top regulators and their main targets

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



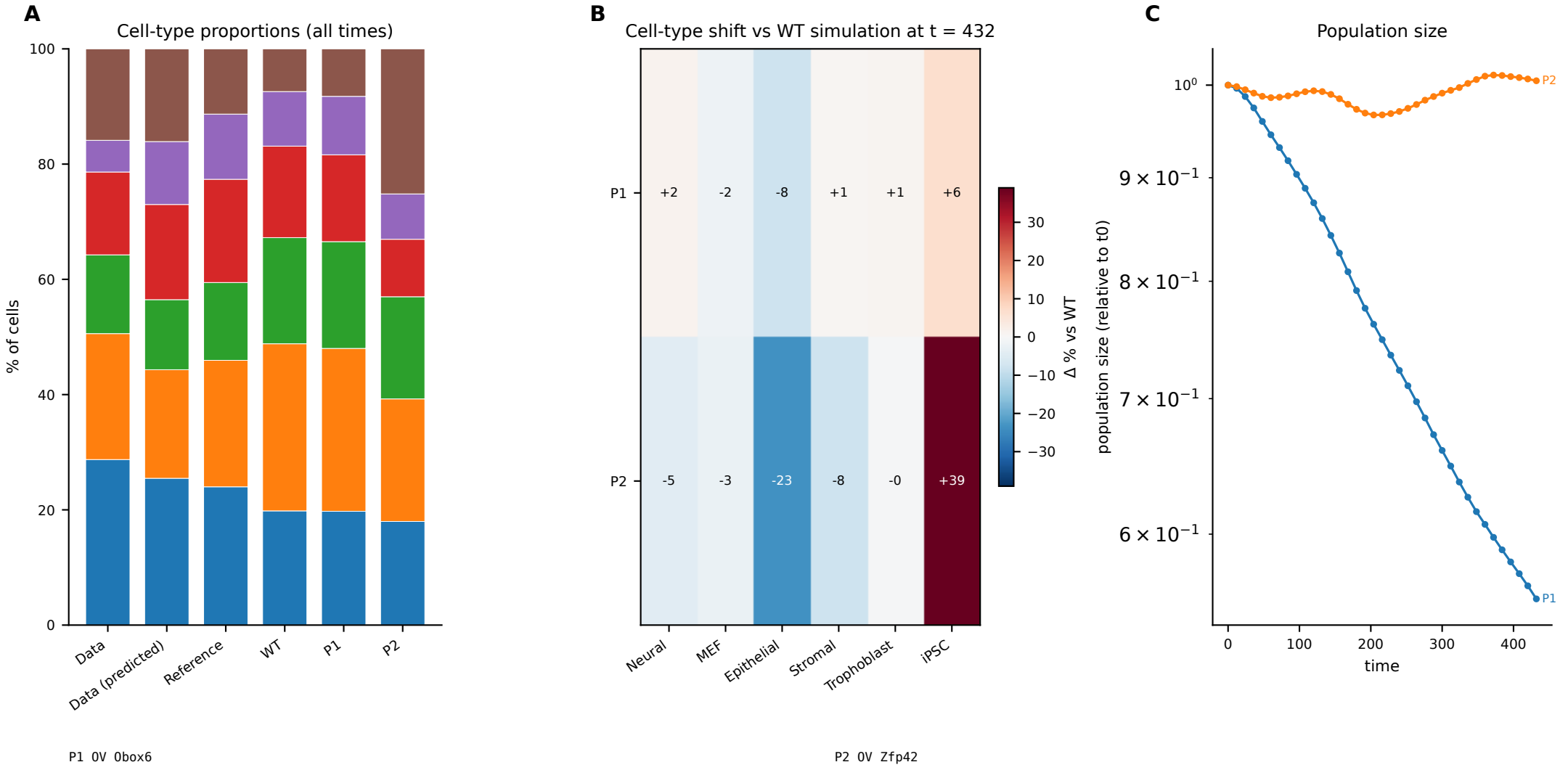
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Perturbed genes (KO/OV) are highlighted in beige.



3. In-silico perturbations — overview

Cell types predicted by per-sample classifiers (logistic) trained on the observed data. Population sizes from the branching simulation (net rates + RATE effects).

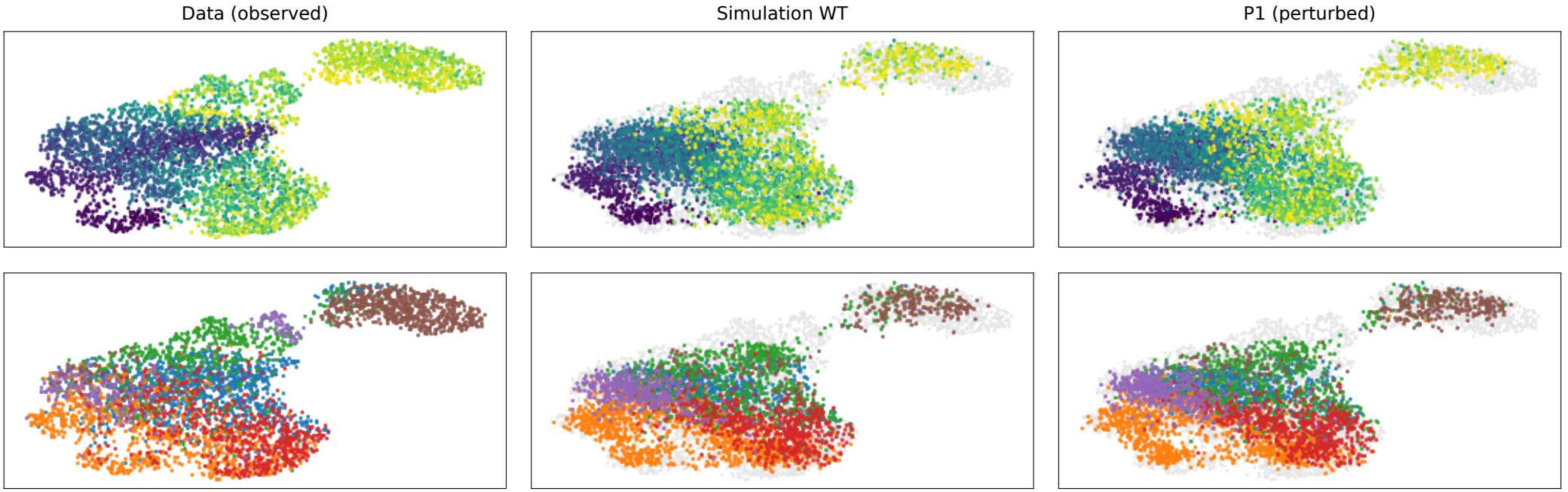


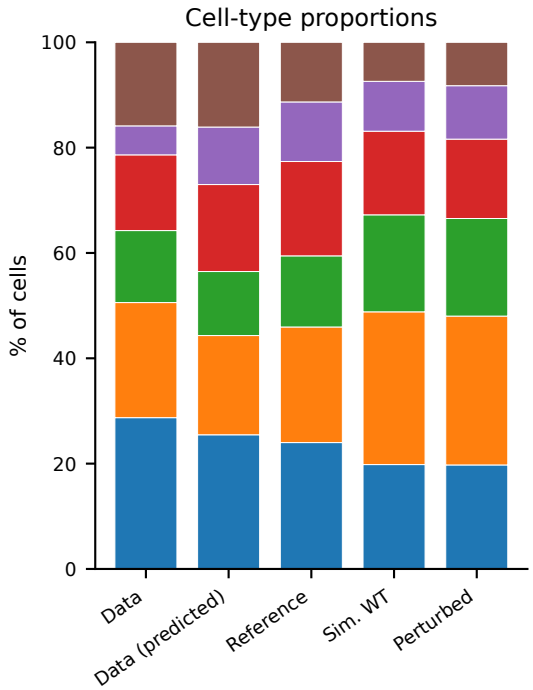
P1 OV Obox6

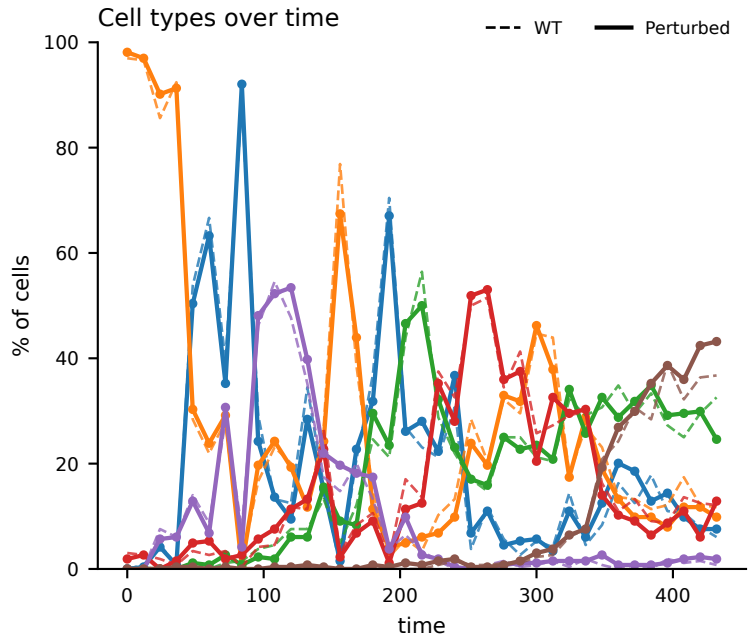
P2 OV Zfp42

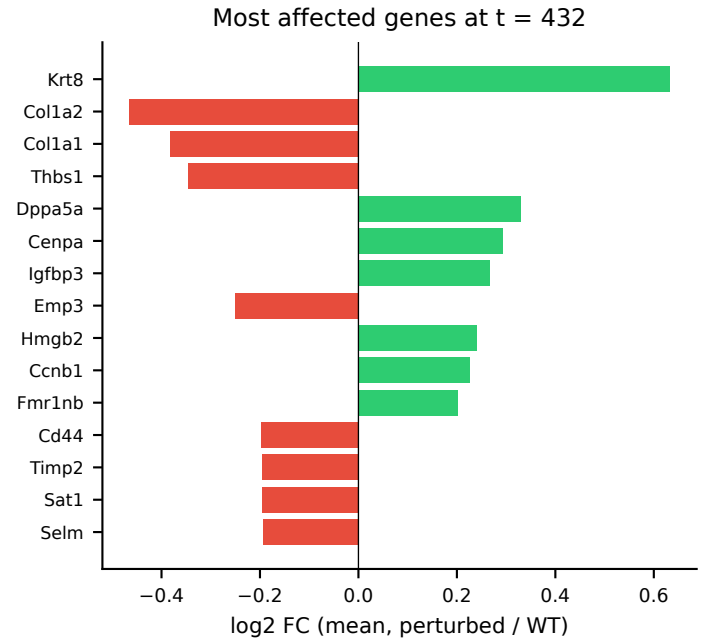
3. Perturbation P1 — OV Obox6

KO_none_OV_Obox6. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).





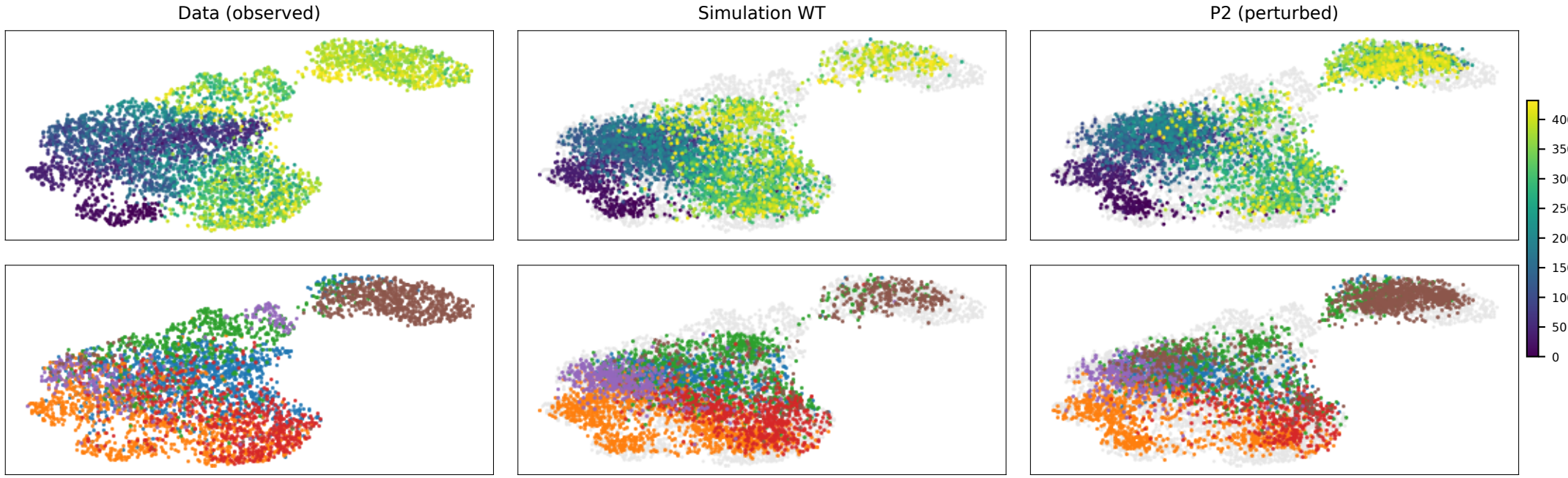


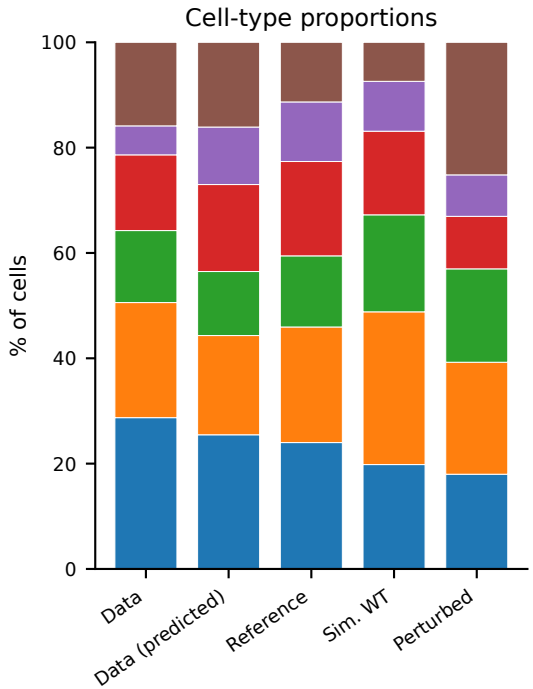


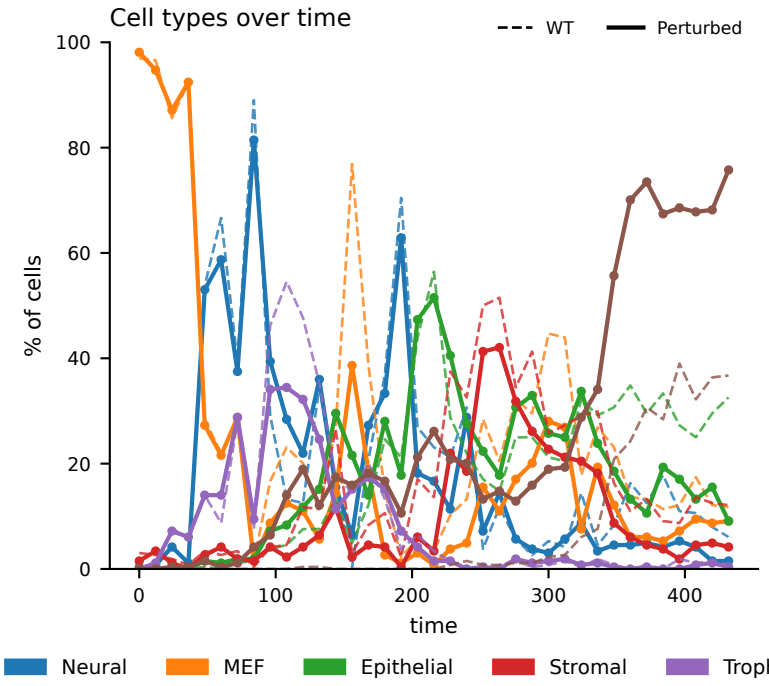


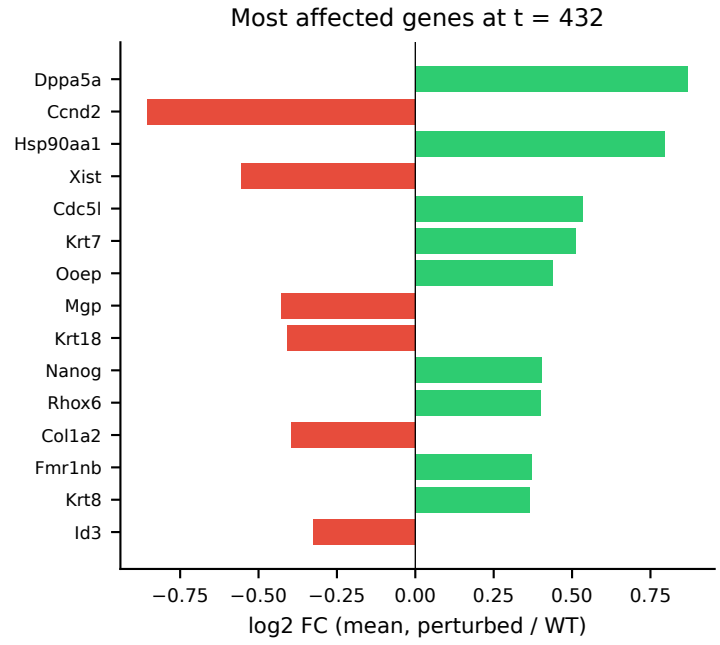
3. Perturbation P2 — OV Zfp42

KO_none_OV_Zfp42. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).











4. Proliferation

Section unavailable:

```
FileNotFoundError('no growth estimate (data_R_opt.npy / data_R_prior.npy from infer_network_structure)')
```

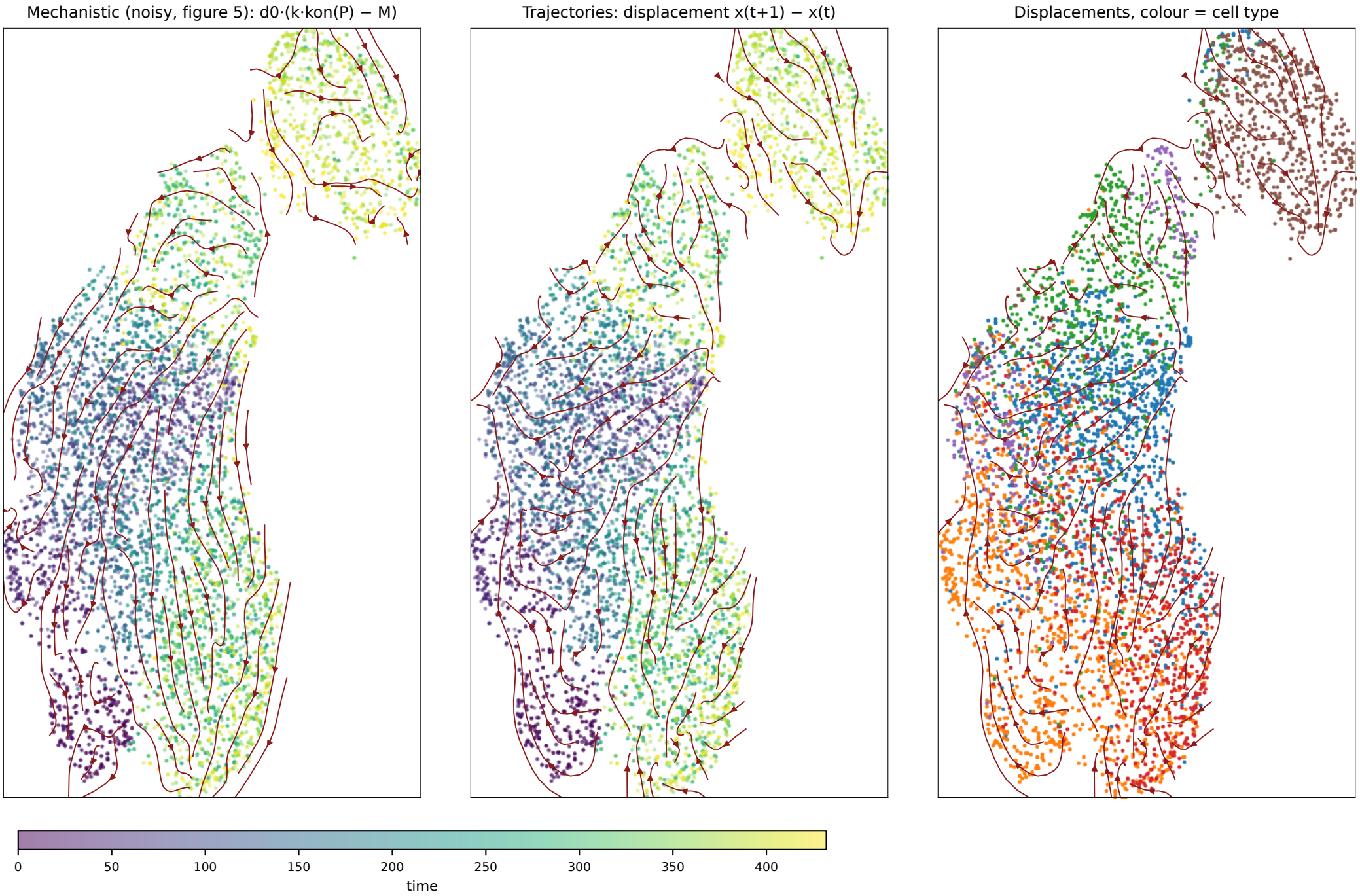
4. Protein dilution

Section unavailable:

```
ValueError('protein_dilution = False, or no birth rate (obs proliferation_birth_rate / net rate)')
```

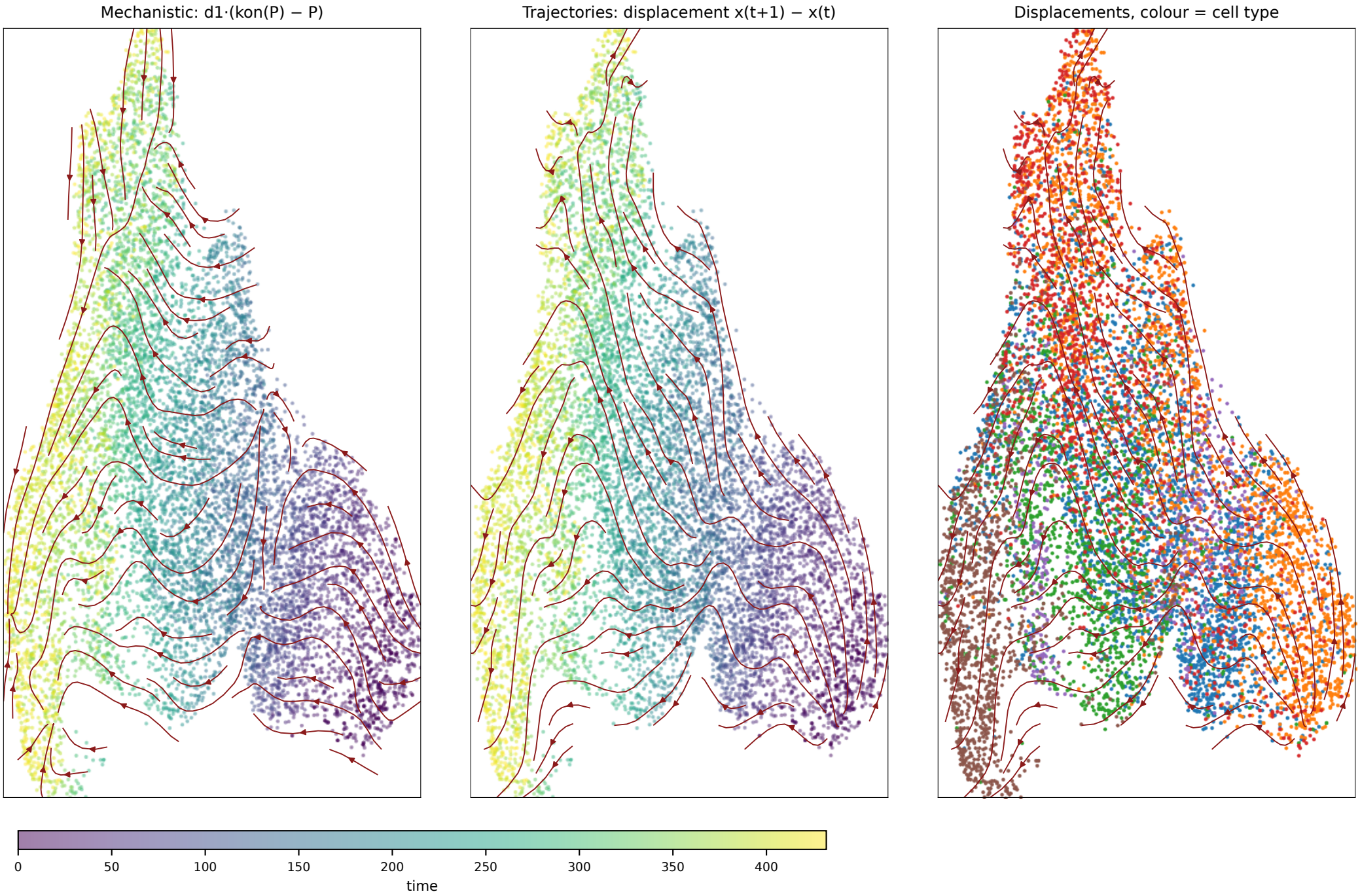
5. Learned dynamics — mRNA velocity and displacement fields

All 9768 trajectory states projected onto the UMAP learned on the observed mRNA (log1p, reference depth); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.06.



5. Learned dynamics — Proteins velocity and displacement fields

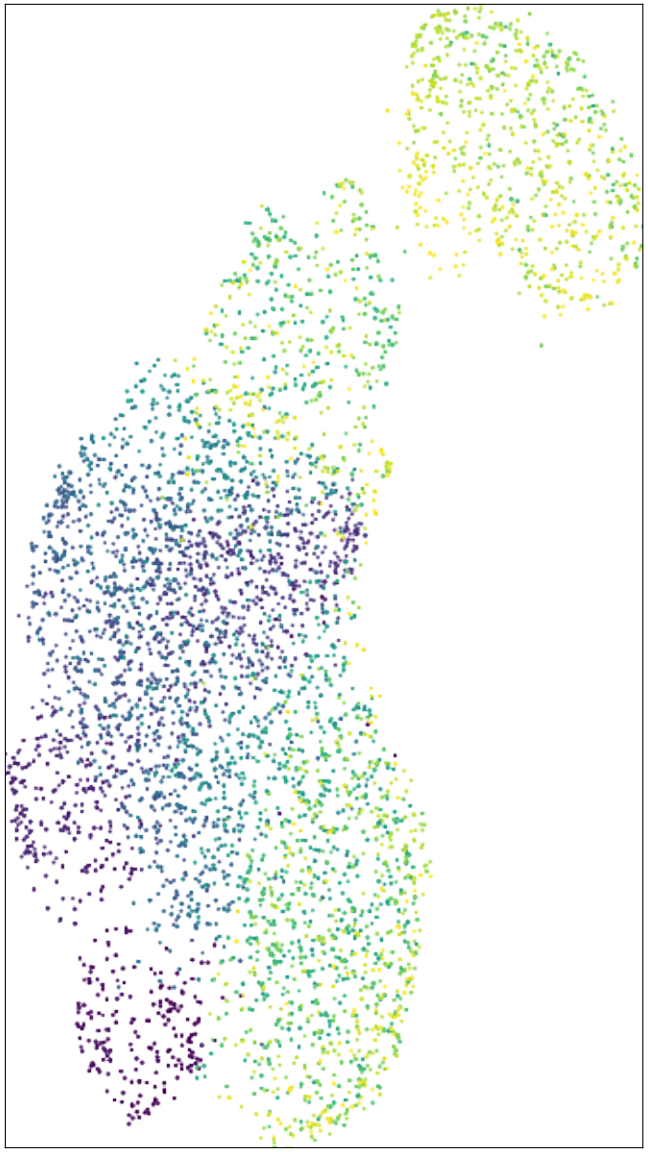
All 9768 trajectory states projected onto the UMAP learned on the protein trajectories (raw levels); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.38.

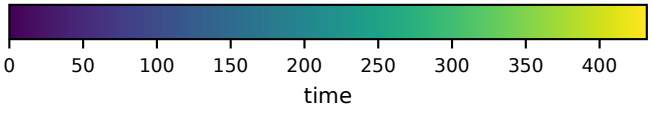


5. Learned dynamics — summary

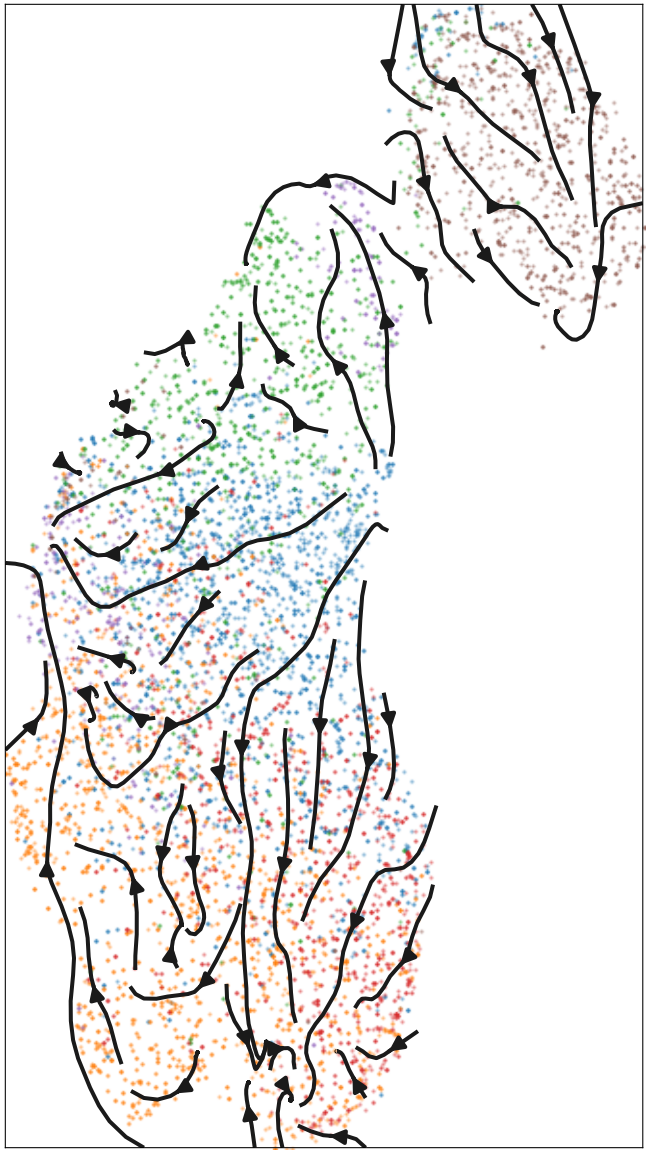
All 9768 trajectory states on the UMAP learned on the observed mRNA (log1p, reference depth); displacement along the inferred trajectories x(t+1) − x(t); background of the field: cell_type.

Time





mRNA displacement along trajectories, cell_type

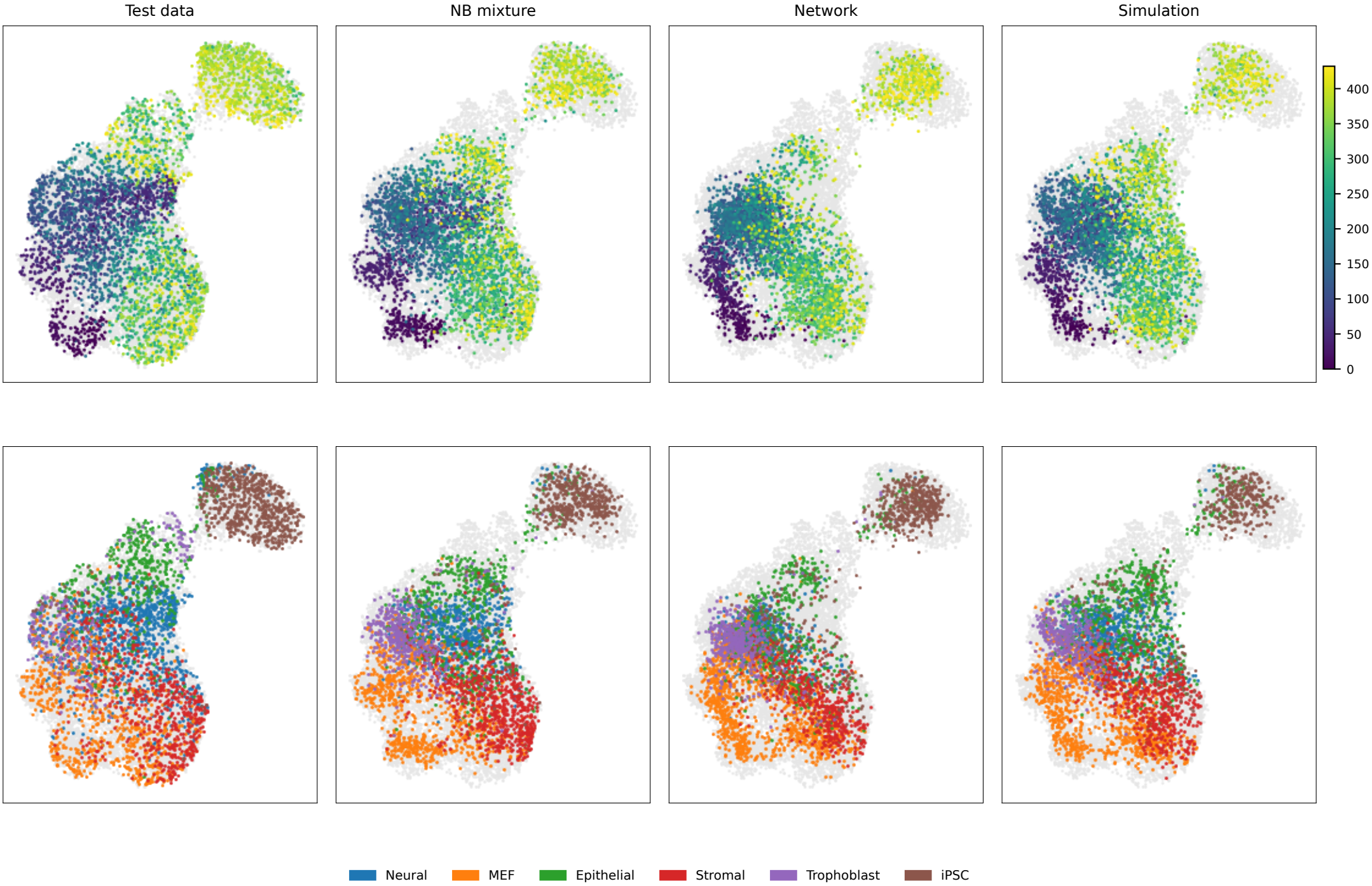


No net proliferation rate



6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the UMAP learned on the observed data (grey: training).



6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

